Commutative Algebra 2

1 Tensoring with A/I: detailed explanation of the four examples

The common principle behind all four examples is the canonical isomorphism

$$(A/I) \otimes_A M \;\cong\; M/IM,$$

where:

- A is a ring,
- $I \subseteq A$ is an ideal,
- M is an A-module,
- IM is the submodule generated by all products im with $i \in I$ and $m \in M$.

Intuitively, tensoring with A/I means that every element of the ideal I is forced to act as 0. In other words, we are "reducing the module M modulo I."

1.1 Element-level meaning of $(A/I) \otimes_A M$

An element of $(A/I) \otimes_A M$ is a finite sum of pure tensors

$$(\overline{a}) \otimes m,$$

where $\overline{a} = a + I \in A/I$ and $m \in M$.

Because $\overline{a} = a \cdot \overline{1}$ in A/I, we have

$$(\overline{a}) \otimes m = (\overline{1}) \otimes am.$$

So every pure tensor can be rewritten in the form

$$\overline{1} \otimes m.$$

Now if $i \in I$, then $\overline{i} = 0$ in A/I, hence

$$\overline{1} \otimes (im) = \overline{i} \otimes m = 0.$$

Therefore every element of IM becomes zero in the tensor product.

This is exactly why the tensor product depends only on the class of m modulo IM. The isomorphism is:

$$\Phi : (A/I) \otimes_A M \longrightarrow M/IM, \qquad (\overline{a}) \otimes m \longmapsto am + IM.$$

Its inverse is given by

$$\Psi : M/IM \longrightarrow (A/I) \otimes_A M, \qquad m + IM \longmapsto \overline{1} \otimes m.$$

So the tensor product remembers precisely the part of M that survives after we impose the rule $I = 0$.

1.2 What survives and what dies

Under the passage from M to $(A/I) \otimes_A M$:

- terms not involving the ideal I survive,
- all terms in the submodule IM vanish.

Thus:

- over $\mathbb{Z}$, tensoring with $\mathbb{Z}/m$ means reducing modulo m,
- over $k[x]$, tensoring with $k[x]/(x)$ means setting $x = 0$.

2 Example 1: $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}^r$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r.$$

By the general formula,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \;\cong\; \mathbb{Z}^r/m\mathbb{Z}^r.$$

Now

$$m\mathbb{Z}^r = \{(ma_1, \dots, ma_r) : a_1, \dots, a_r \in \mathbb{Z}\}.$$

So $\mathbb{Z}^r/m\mathbb{Z}^r$ is obtained by reducing each coordinate modulo m. Therefore

$$\mathbb{Z}^r/m\mathbb{Z}^r \;\cong\; (\mathbb{Z}/m)^r.$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \;\cong\; (\mathbb{Z}/m)^r.$$

2.1 Element-wise description

Let $e_1, \dots, e_r$ be the standard basis of $\mathbb{Z}^r$. Any vector $v \in \mathbb{Z}^r$ can be written as

$$v = a_1e_1 + \cdots + a_re_r.$$

Then

$$[1]_m \otimes v = [1]_m \otimes (a_1e_1 + \cdots + a_re_r) = [a_1]_m \otimes e_1 + \cdots + [a_r]_m \otimes e_r.$$

So the tensor product simply remembers the coefficients modulo m.

2.2 Concrete example

Take $m = 5$, $r = 2$, and $v = (7, 11)$. Then

$$(7, 11) \equiv (2, 1) \pmod 5.$$

Thus

$$[1]_5 \otimes (7, 11) \quad \text{corresponds to} \quad ([2]_5, [1]_5) \in (\mathbb{Z}/5)^2.$$

2.3 Interpretation

The "vectors" here are genuine integer vectors in $\mathbb{Z}^r$, but after tensoring with $\mathbb{Z}/m$, only their coordinates modulo m remain visible.

3 Example 2: $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}/n$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n).$$

Using the same principle,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \;\cong\; (\mathbb{Z}/n)/m(\mathbb{Z}/n).$$

So we must understand the subgroup $m(\mathbb{Z}/n)$.

3.1 Computing $m(\mathbb{Z}/n)$

Inside $\mathbb{Z}/n$, multiplication by m gives the subgroup

$$m(\mathbb{Z}/n) = \{[mk]_n : k \in \mathbb{Z}\}.$$

This is the cyclic subgroup generated by $[m]_n$.

The order of $[m]_n$ in $\mathbb{Z}/n$ is

$$\frac{n}{\gcd(m,n)},$$

so the subgroup $m(\mathbb{Z}/n)$ has exactly that many elements.

Therefore the quotient $(\mathbb{Z}/n)/m(\mathbb{Z}/n)$ has cardinality

$$\frac{n}{n/\gcd(m,n)} = \gcd(m,n).$$

Since the quotient of a cyclic group is cyclic, we get

$$(\mathbb{Z}/n)/m(\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n).$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \;\cong\; \mathbb{Z}/\gcd(m,n).$$

3.2 Element-wise description

The tensor product is generated by the single element

$$[1]_m \otimes [1]_n.$$

Indeed, for any $a, b \in \mathbb{Z}$,

$$[a]_m \otimes [b]_n = ab\big([1]_m \otimes [1]_n\big).$$

So the whole tensor product is cyclic.

Moreover,

$$m\big([1]_m \otimes [1]_n\big) = 0 \qquad \text{and} \qquad n\big([1]_m \otimes [1]_n\big) = 0,$$

so the order of the generator divides both m and n, hence divides $\gcd(m,n)$. In fact, its exact order is $\gcd(m,n)$.

3.3 Concrete example

Let $m = 12$ and $n = 18$. Then

$$\gcd(12, 18) = 6,$$

so

$$(\mathbb{Z}/12) \otimes_{\mathbb{Z}} (\mathbb{Z}/18) \cong \mathbb{Z}/6.$$

The element

$$[1]_{12} \otimes [1]_{18}$$

is a generator of this cyclic group.

3.4 Interpretation

This tensor product retains only the *common torsion* visible in both cyclic modules:

- $\mathbb{Z}/m$ has m-torsion,
- $\mathbb{Z}/n$ has n-torsion,
- their tensor product detects the part compatible with both, namely $\gcd(m, n)$-torsion.

4 Example 3: $A = k[x]$, $I = (x)$ — evaluation at $x = 0$

Now let

$$A = k[x], \qquad I = (x).$$

Then

$$A/I = k[x]/(x).$$

But quotienting by (x) kills all multiples of x, so every polynomial becomes equal to its constant term. Thus

$$k[x]/(x) \cong k,$$

via the map

$$f(x) \longmapsto f(0).$$

So in this situation,

$$A/I \cong k.$$

Therefore for any A-module M,

$$k \otimes_{k[x]} M \;\cong\; M/xM.$$

4.1 Why this means "set $x = 0$"

Since $x = 0$ in the quotient ring $k[x]/(x)$, we have in the tensor product:

$$1 \otimes (xm) = x \otimes m = 0.$$

So every element of the form xm vanishes. That is why we divide by xM.

This is exactly the module-theoretic meaning of evaluating at $x = 0$: all x-terms disappear.

4.2 Particular case: $M = k[x]$

If $M = A = k[x]$, then
$$k \otimes_{k[x]} k[x] \;\cong\; k[x]/xk[x].$$
But $xk[x]$ is the set of polynomials with zero constant term, so the quotient keeps only the constant term:
$$k[x]/xk[x] \cong k.$$

Thus
$$k \otimes_{k[x]} k[x] \cong k.$$

4.3 Element-wise example

Take the polynomial
$$f(x) = 3 + 2x + x^2.$$
Then in the tensor product,
$$1 \otimes f(x) = 1 \otimes (3 + 2x + x^2).$$
Since all terms involving x vanish,
$$1 \otimes (3 + 2x + x^2) = 1 \otimes 3.$$
So only the constant term remains.

4.4 Interpretation

Tensoring with $k \cong k[x]/(x)$ means:

- forget all multiples of x,
- keep only what survives at $x = 0$,
- algebraically: pass from M to the quotient M/xM.

5 Example 4: $A = k[x]$, $I = (x)$, $M = A/(x^2)$

Now let
$$A = k[x], \qquad I = (x), \qquad M = A/(x^2) = k[x]/(x^2).$$
We want to compute
$$(A/I) \otimes_A M = k \otimes_{k[x]} A/(x^2).$$

By the general rule,
$$(A/I) \otimes_A M \;\cong\; M/xM.$$

So we must compute xM.

5.1 Structure of $M = A/(x^2)$

In the quotient $k[x]/(x^2)$, every polynomial is equivalent to one of the form

$$a + bx \pmod{x^2}.$$

So as a k-vector space,

$$M = k \cdot 1 \oplus k \cdot x.$$

A convenient basis is

$$\{1, x\}.$$

However, inside this module we have

$$x^2 = 0.$$

So x is not zero, but it is nilpotent.

5.2 Computing xM

Take an arbitrary element

$$a + bx \in M.$$

Multiplying by x, we get

$$x(a + bx) = ax + bx^2.$$

Since $x^2 = 0$ in M, this becomes

$$x(a + bx) = ax.$$

Therefore

$$xM = kx.$$

So the quotient is

$$M/xM = (k \cdot 1 \oplus k \cdot x)/(k \cdot x) \cong k.$$

Hence

$$k \otimes_{k[x]} A/(x^2) \;\cong\; M/xM \;\cong\; k.$$

5.3 Element-wise tensor computation

Take the class of $a + bx$ in $M = A/(x^2)$. Then in the tensor product,

$$1 \otimes (a + bx) = 1 \otimes a + 1 \otimes bx.$$

But

$$1 \otimes bx = b(1 \otimes x).$$

Now $x = 0$ in $k = A/(x)$, so

$$1 \otimes x = x \otimes 1 = 0.$$

Therefore

$$1 \otimes (a + bx) = 1 \otimes a.$$

So again, only the constant part survives.

5.4 Interpretation

This example is important because $M = A/(x^2)$ is larger than k: it has an extra "infinitesimal" direction kx. But after tensoring with $k = A/(x)$, the x-part disappears.

So even though $x \neq 0$ in M, it becomes invisible after imposing $x = 0$.

6 What are the "vectors" or "elements" in these examples?

These are not always vectors in the geometric sense. They are simply *elements of modules.*

- In Example 1, the elements are tuples

$$(a_1, \ldots, a_r) \in \mathbb{Z}^r.$$

  These behave like integer vectors.

- In Example 2, the elements are residue classes

$$[a]_n \in \mathbb{Z}/n.$$

- In Example 3, the elements are arbitrary module elements $m \in M$, where M is any $k[x]$-module.

- In Example 4, the elements are classes

$$a + bx \pmod{x^2},$$

  so each element has a constant part and an x-part.

In all cases, after tensoring with A/I, every element can be represented by something of the form

$$\bar{1} \otimes m,$$

and two elements become equal precisely when they differ by something from IM.

Thus the tensor product is measuring the module M after "forgetting" everything killed by the ideal I.

7 One-line intuition for each example

- **Example 1:** reduce each coordinate modulo m.
- **Example 2:** keep only the common torsion shared by $\mathbb{Z}/m$ and $\mathbb{Z}/n$.
- **Example 3:** set $x = 0$ in the module.
- **Example 4:** set $x = 0$ even in a module where $x \neq 0$ but $x^2 = 0$, so the nilpotent x-part disappears.

8 Final universal picture

The construction

$$(A/I) \otimes_A M$$

means that we are viewing the module M after passing from the ring A to the quotient ring A/I.

Equivalently:

- coefficients are reduced modulo I,
- every element of I acts as zero,
- all elements of IM vanish,
- the remaining information is exactly the quotient module M/IM.

So the identity

$$(A/I) \otimes_A M \;\cong\; M/IM$$

is the precise algebraic statement that tensoring with A/I is the same as reducing the module M modulo the action of the ideal I.

Problem 5 — Zero divisors, localization, and associated primes via $D(A)$

Exact statement. In a ring A, let $D(A)$ denote the set of prime ideals $\mathfrak{p}$ which satisfy the following condition: there exists an $a \in A$ such that $\mathfrak{p}$ is minimal in $V(\operatorname{Ann}(a))$.

(a) Show that x is a zero divisor in A if and only if $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$.

(b) Let S be a multiplicatively closed subset of A. Under the canonical identification of $\operatorname{Spec} S^{-1}A$ with a subset of $\operatorname{Spec} A$, show that

$$D(S^{-1}A) = D(A) \cap \operatorname{Spec} S^{-1}A.$$

(c) If the 0 ideal has a primary decomposition, show that $D(A)$ is the set of associated primes of A.

Solution.
(a) First suppose that x is a zero divisor. Then there exists $a \neq 0$ such that

$$xa = 0.$$

Hence $x \in \operatorname{Ann}(a)$. Choose a prime ideal $\mathfrak{p}$ minimal over $\operatorname{Ann}(a)$. Then $\mathfrak{p} \in D(A)$ by definition, and $\operatorname{Ann}(a) \subseteq \mathfrak{p}$, so $x \in \mathfrak{p}$.

Conversely, suppose $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$. Then there exists $a \in A$ such that $\mathfrak{p}$ is minimal over $\operatorname{Ann}(a)$. Let

$$I = \operatorname{Ann}(a), \qquad B = A/I.$$

Then $\mathfrak{q} := \mathfrak{p}/I$ is a minimal prime of B, and $\bar{x} \in \mathfrak{q}$.

Localizing at $\mathfrak{q}$, the ring $B_\mathfrak{q}$ has only one prime ideal, namely $\mathfrak{q}B_\mathfrak{q}$. Hence every element of $\mathfrak{q}B_\mathfrak{q}$ is nilpotent. In particular, $\bar{x}/1$ is nilpotent in $B_\mathfrak{q}$. Thus there exist $s \notin \mathfrak{q}$ and $n \geq 1$ such that

$$s\bar{x}^n = 0 \quad \text{in } B.$$

Choose n minimal with this property. Then

$$\bar{b} := s\bar{x}^{n-1} \neq 0, \qquad \bar{x}\,\bar{b} = 0.$$

So $\bar{x}$ is a zero divisor in B. Lift $\bar{b}$ to some $b \in A$. Since $\bar{b} \neq 0$, we have $b \notin I = \operatorname{Ann}(a)$, so $ba \neq 0$. Also $\bar{x}\,\bar{b} = 0$ means $xb \in I$, hence

$$x(ba) = (xb)a = 0.$$

Therefore x is a zero divisor in A.

Thus

$$x \text{ is a zero divisor in } A \iff x \in \mathfrak{p} \text{ for some } \mathfrak{p} \in D(A).$$

(b) Identify $\operatorname{Spec} S^{-1}A$ with the set of prime ideals $\mathfrak{p} \subset A$ such that $\mathfrak{p} \cap S = \varnothing$.

First let $\mathfrak{q} \in D(S^{-1}A)$. Then there exists $a/s \in S^{-1}A$ such that $\mathfrak{q}$ is minimal over

$$\operatorname{Ann}_{S^{-1}A}(a/s) = S^{-1}\operatorname{Ann}_A(a).$$

Write $\mathfrak{q} = S^{-1}\mathfrak{p}$, where $\mathfrak{p} \cap S = \varnothing$. We claim that $\mathfrak{p}$ is minimal over $\operatorname{Ann}_A(a)$. Indeed, if $\mathfrak{r} \subsetneq \mathfrak{p}$ is a prime ideal containing $\operatorname{Ann}_A(a)$, then $\mathfrak{r} \cap S = \varnothing$ because $\mathfrak{r} \subseteq \mathfrak{p}$. Hence $S^{-1}\mathfrak{r}$ is a prime of $S^{-1}A$ properly contained in $\mathfrak{q}$ and containing $S^{-1}\operatorname{Ann}_A(a)$, contradiction. So $\mathfrak{p} \in D(A)$, and $\mathfrak{p} \in \operatorname{Spec} S^{-1}A$. Therefore

$$D(S^{-1}A) \subseteq D(A) \cap \operatorname{Spec} S^{-1}A.$$

Conversely, let $\mathfrak{p} \in D(A) \cap \operatorname{Spec} S^{-1}A$. Then $\mathfrak{p} \cap S = \varnothing$, and there exists $a \in A$ such that $\mathfrak{p}$ is minimal over $\operatorname{Ann}_A(a)$. Since $\operatorname{Ann}_A(a) \subseteq \mathfrak{p}$ and $\mathfrak{p} \cap S = \varnothing$, the element $a/1$ is nonzero in $S^{-1}A$. Also

$$\operatorname{Ann}_{S^{-1}A}(a/1) = S^{-1}\operatorname{Ann}_A(a).$$

If $S^{-1}\mathfrak{r} \subsetneq S^{-1}\mathfrak{p}$ is a prime containing $S^{-1}\operatorname{Ann}_A(a)$, then $\mathfrak{r} \subsetneq \mathfrak{p}$ is a prime containing $\operatorname{Ann}_A(a)$, contradiction. Hence $S^{-1}\mathfrak{p} \in D(S^{-1}A)$. Thus

$$D(A) \cap \operatorname{Spec} S^{-1}A \subseteq D(S^{-1}A).$$

Therefore

$$D(S^{-1}A) = D(A) \cap \operatorname{Spec} S^{-1}A.$$

(c) Assume that

$$0 = Q_1 \cap \cdots \cap Q_n$$

is an irredundant primary decomposition, where each Q_i is $\mathfrak{p}_i$-primary. Then

$$\operatorname{Ass}_A(A) = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}.$$

We show that $D(A)$ is the same set.

First, let $\mathfrak{p}_i$ be one of these primes. Since the decomposition is irredundant, there exists

$$a_i \in \bigcap_{j \neq i} Q_j \setminus Q_i.$$

Then

$$\mathrm{Ann}(a_i) = \{r \in A : ra_i = 0\}.$$

Because $a_i \in Q_j$ for all $j \neq i$, the condition $ra_i = 0$ is equivalent to $ra_i \in Q_i$. Hence

$$\mathrm{Ann}(a_i) = (Q_i : a_i).$$

Since Q_i is $\mathfrak{p}_i$-primary and $a_i \notin Q_i$, the colon ideal $(Q_i : a_i)$ is $\mathfrak{p}_i$-primary. Therefore

$$\sqrt{\mathrm{Ann}(a_i)} = \mathfrak{p}_i,$$

so $\mathfrak{p}_i$ is the unique minimal prime over $\mathrm{Ann}(a_i)$. Hence $\mathfrak{p}_i \in D(A)$.

Thus

$$\mathrm{Ass}_A(A) \subseteq D(A).$$

Conversely, let $\mathfrak{p} \in D(A)$. Then $\mathfrak{p}$ is minimal over $\mathrm{Ann}(a)$ for some $a \in A$. Since

$$0 = \bigcap_{i=1}^{n} Q_i,$$

we obtain

$$\mathrm{Ann}(a) = \bigcap_{i=1}^{n} (Q_i : a).$$

For each i, either $a \in Q_i$, in which case $(Q_i : a) = A$, or $a \notin Q_i$, in which case $(Q_i : a)$ is $\mathfrak{p}_i$-primary. Therefore $\mathrm{Ann}(a)$ has a primary decomposition whose associated primes are among the $\mathfrak{p}_i$. Hence every minimal prime over $\mathrm{Ann}(a)$, in particular $\mathfrak{p}$, must equal one of the $\mathfrak{p}_i$. So

$$D(A) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\} = \mathrm{Ass}_A(A).$$

Therefore

$$D(A) = \mathrm{Ass}_A(A).$$

Problem 6 — The ideals $S_{\mathfrak{p}}(0)$ and minimal primes

Exact statement. For $\mathfrak{p} \subset A$ a prime ideal, denote by $S_{\mathfrak{p}}(0)$ the kernel of the canonical homomorphism

$$A \to A_{\mathfrak{p}}.$$

Prove:

(a) $S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}$.

(b) If $\mathfrak{p} \supseteq \mathfrak{p}'$, then $S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0)$.

(c) $\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}$ if and only if $\mathfrak{p}$ is a minimal prime ideal of A.

(d) $\bigcap_{\mathfrak{p}\in D(A)} S_{\mathfrak{p}}(0) = 0$, where $D(A)$ is the set defined in the previous exercise.

Solution.
Recall that
$$S_{\mathfrak{p}}(0) = \{a \in A : \exists s \notin \mathfrak{p} \text{ such that } sa = 0\}.$$

(a) Let $a \in S_{\mathfrak{p}}(0)$. Then $sa = 0$ for some $s \notin \mathfrak{p}$. Since $0 \in \mathfrak{p}$, we have $sa \in \mathfrak{p}$. Because $\mathfrak{p}$ is prime and $s \notin \mathfrak{p}$, it follows that $a \in \mathfrak{p}$. Hence

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}.$$

(b) Suppose $\mathfrak{p} \supseteq \mathfrak{p}'$, and let $a \in S_{\mathfrak{p}}(0)$. Then there exists $s \notin \mathfrak{p}$ such that $sa = 0$. Since $\mathfrak{p}' \subseteq \mathfrak{p}$, we also have $s \notin \mathfrak{p}'$. Thus $a \in S_{\mathfrak{p}'}(0)$. Therefore

$$S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0).$$

(c) First assume that $\mathfrak{p}$ is a minimal prime ideal of A. By part (a),

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{p},$$

so

$$\sqrt{S_{\mathfrak{p}}(0)} \subseteq \mathfrak{p}.$$

Now let $x \in \mathfrak{p}$. In the localized ring $A_{\mathfrak{p}}$, the prime ideals correspond to prime ideals of A contained in $\mathfrak{p}$. Since $\mathfrak{p}$ is minimal, the only prime ideal of $A_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$. Therefore the nilradical of $A_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$, so $x/1$ is nilpotent in $A_{\mathfrak{p}}$. Hence for some $n \geq 1$,

$$(x/1)^n = 0,$$

which means $x^n \in S_{\mathfrak{p}}(0)$. Thus $x \in \sqrt{S_{\mathfrak{p}}(0)}$, and so

$$\mathfrak{p} \subseteq \sqrt{S_{\mathfrak{p}}(0)}.$$

Therefore

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}.$$

Conversely, suppose

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}.$$

If $\mathfrak{p}$ were not minimal, there would exist a prime ideal $\mathfrak{q} \subsetneq \mathfrak{p}$. We claim that

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{q}.$$

Indeed, if $a \in S_{\mathfrak{p}}(0)$, then $sa = 0$ for some $s \notin \mathfrak{p}$. Since $\mathfrak{q} \subseteq \mathfrak{p}$, also $s \notin \mathfrak{q}$. Because $\mathfrak{q}$ is prime and $sa = 0 \in \mathfrak{q}$, it follows that $a \in \mathfrak{q}$. Therefore

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{q},$$

hence

$$\sqrt{S_{\mathfrak{p}}(0)} \subseteq \mathfrak{q} \subsetneq \mathfrak{p},$$

contradiction. Thus $\mathfrak{p}$ must be minimal.

So

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p} \iff \mathfrak{p} \text{ is a minimal prime ideal of } A.$$

(d) We prove that

$$\bigcap_{\mathfrak{p}\in D(A)} S_{\mathfrak{p}}(0) = 0.$$

Let $x \neq 0$. Choose a prime ideal $\mathfrak{p}$ minimal over $\operatorname{Ann}(x)$. Then $\mathfrak{p} \in D(A)$ by definition. We claim that $x \notin S_{\mathfrak{p}}(0)$. Otherwise there would exist $s \notin \mathfrak{p}$ such that

$$sx = 0.$$

Then $s \in \operatorname{Ann}(x) \subseteq \mathfrak{p}$, contradiction. Hence $x \notin S_{\mathfrak{p}}(0)$.

Therefore no nonzero element belongs to every $S_{\mathfrak{p}}(0)$, and we conclude that

$$\bigcap_{\mathfrak{p}\in D(A)} S_{\mathfrak{p}}(0) = 0.$$

Problem 7 — A local-global criterion for Noetherianity

Exact statement. Let A be a ring such that

(1) for every maximal ideal $\mathfrak{m}$ of A, the local ring $A_{\mathfrak{m}}$ is Noetherian;

(2) for each non-zero $x \in A$, the set of maximal ideals of A which contain x is finite.

Show that A is Noetherian.

Solution.

We show that every ideal of A is finitely generated.

Let $I \subseteq A$ be an ideal. If $I = 0$, there is nothing to prove. So assume $I \neq 0$, and choose $0 \neq x \in I$.

By assumption (2), only finitely many maximal ideals contain x. Denote them by

$$\mathfrak{m}_1, \dots, \mathfrak{m}_r.$$

For each i, the ideal $I_{\mathfrak{m}_i}$ of the Noetherian ring $A_{\mathfrak{m}_i}$ is finitely generated. Hence there exist elements

$$a_{i1}, \dots, a_{in_i} \in I$$

such that

$$I_{\mathfrak{m}_i} = (a_{i1}, \dots, a_{in_i})_{\mathfrak{m}_i}.$$

Now define

$$J = (x, a_{11}, \dots, a_{1n_1}, \dots, a_{r1}, \dots, a_{rn_r}) \subseteq I.$$

This is a finitely generated ideal. We claim that $I = J$.

Let $\mathfrak{m}$ be any maximal ideal of A.

If $x \notin \mathfrak{m}$, then $x/1$ is a unit in $A_{\mathfrak{m}}$. Since $x \in J \subseteq I$, both $J_{\mathfrak{m}}$ and $I_{\mathfrak{m}}$ contain a unit, hence

$$J_{\mathfrak{m}} = A_{\mathfrak{m}} = I_{\mathfrak{m}}.$$

If $x \in \mathfrak{m}$, then $\mathfrak{m} = \mathfrak{m}_i$ for some i, and by construction

$$J_{\mathfrak{m}} = I_{\mathfrak{m}}.$$

Thus

$$J_{\mathfrak{m}} = I_{\mathfrak{m}} \qquad \text{for every maximal ideal } \mathfrak{m}.$$

Consider the quotient module I/J. For every maximal ideal $\mathfrak{m}$,

$$(I/J)_{\mathfrak{m}} = 0.$$

A module is zero if and only if all its localizations at maximal ideals are zero. Therefore $I/J = 0$, so $I = J$.

Hence every ideal of A is finitely generated. Therefore A is Noetherian.

Theory: ideals, quotients, coordinate rings, and localization on concrete polynomial examples

Throughout this section, let k be a field. We will work mainly in the polynomial rings

$$k[x], \qquad k[x, y].$$

The goal is not only to state abstract definitions, but to understand what these notions look like on explicit polynomials, what their elements are, and what they mean geometrically.

1. The ambient polynomial ring

Take

$$R = k[x, y].$$

An element of R is an ordinary polynomial in two variables, for example

$$f(x, y) = x^3 + 2x^2y - 5y + 7.$$

The ring $k[x, y]$ is the algebraic ambient space of all polynomial expressions in the coordinates x and y.

Geometrically, one thinks of:

- x and y as coordinate functions on the affine plane k^2,
- a polynomial equation $f(x, y) = 0$ as defining a curve or algebraic subset of the plane.

Thus $k[x, y]$ is the algebraic side of affine plane geometry.

2. Ideals: concrete algebraic meaning

An ideal $I \subseteq k[x, y]$ is a subset such that:

- if $f, g \in I$, then $f - g \in I$,
- if $f \in I$ and $h \in k[x, y]$, then $hf \in I$.

So an ideal is closed under addition and under multiplication by arbitrary polynomials.

The geometric idea is fundamental:

If $I \subseteq k[x,y]$, then its common zero set is

$$V(I) = \{(a,b) \in k^2 : f(a,b) = 0 \text{ for every } f \in I\}.$$

Thus:

- an ideal is a collection of polynomial relations,
- the set $V(I)$ is the geometric object cut out by those relations.

3. Principal ideals: one equation

Take the ideal

$$I = (y - x^2) \subseteq k[x,y].$$

This means

$$(y - x^2) = \{h(x,y)(y - x^2) : h(x,y) \in k[x,y]\}.$$

What are its elements? Every element is a multiple of $y - x^2$. For example:

$$x(y - x^2) = xy - x^3,$$

$$(y+1)(y - x^2) = y^2 + y - x^2y - x^2.$$

All such polynomials vanish on every point satisfying $y = x^2$.

Geometry. The equation

$$y - x^2 = 0$$

defines the parabola

$$V(y - x^2) = \{(a, a^2) : a \in k\}.$$

Quotient ring. Now form

$$k[x,y]/(y - x^2).$$

This means that in the quotient ring we impose the relation

$$y = x^2.$$

So every polynomial in x and y can be simplified by replacing y with x^2.

Examples:

$$x^2 + y \equiv x^2 + x^2 = 2x^2,$$

$$y^2 + x \equiv x^4 + x,$$

$$xy + y^3 \equiv x \cdot x^2 + x^6 = x^3 + x^6.$$

Hence there is an isomorphism

$$k[x,y]/(y - x^2) \cong k[x].$$

Coordinate ring meaning. This quotient is the coordinate ring of the parabola. Its elements are polynomial functions on the parabola.

For example, the class of y is equal to the class of x^2, because on the parabola the two functions agree:

$$(a, a^2) \mapsto a^2.$$

Thus:

- in the ambient plane k^2, the polynomials y and x^2 are different,
- on the parabola, they define the same function.

That is exactly what quotienting by $(y - x^2)$ expresses.

4. Ideals generated by several polynomials

Take

$$I = (x, y) \subseteq k[x, y].$$

This consists of all polynomials of the form

$$xf(x, y) + yg(x, y), \qquad f, g \in k[x, y].$$

What polynomials lie in (x, y)? Exactly those with zero constant term.

Indeed:

- every polynomial of the form $xf + yg$ vanishes at $(0, 0)$,
- conversely, every polynomial with zero constant term can be written as such a combination.

For example:

$$x^2 + xy + y^3 \in (x, y),$$
$$x + y + 5xy^2 \in (x, y),$$

but

$$1 + x \notin (x, y)$$

because it has nonzero constant term.

Geometry. The zero set is

$$V(x, y) = \{(0, 0)\}.$$

So the ideal (x, y) cuts out the origin.

Quotient. In the quotient

$$k[x, y]/(x, y),$$

both x and y become 0, so every polynomial reduces to its constant term. Therefore

$$k[x, y]/(x, y) \cong k.$$

This is the coordinate ring of a single point.

Interpretation. This is the basic prototype of a maximal ideal: quotienting by it gives the field k, and geometrically it corresponds to one point.

5. Maximal ideals: points in affine space

In $k[x, y]$, when k is algebraically closed, the maximal ideals are exactly

$$(x - a,\ y - b), \qquad (a, b) \in k^2.$$

Take for example

$$\mathfrak{m} = (x - 2,\ y + 3).$$

Geometry. Its zero set is the single point

$$(2, -3).$$

Quotient. In the quotient

$$k[x, y]/(x - 2, y + 3),$$

we have

$$x \equiv 2, \qquad y \equiv -3.$$

So every polynomial reduces to the scalar obtained by evaluation at that point. For example:

$$x^2 + y \equiv 4 - 3 = 1,$$
$$x^3 - 2y \equiv 8 - 2(-3) = 14.$$

Hence

$$k[x, y]/(x - 2, y + 3) \cong k.$$

Meaning. Quotienting by a maximal ideal means evaluating polynomial functions at a point. Thus maximal ideals correspond to points.

6. Prime ideals: irreducible geometry

Take

$$\mathfrak{p} = (y) \subseteq k[x, y].$$

Quotient. We have

$$k[x, y]/(y) \cong k[x].$$

Since $k[x]$ is an integral domain, it follows that (y) is a prime ideal.

Geometry. The zero set is

$$V(y) = \{(a, 0) : a \in k\},$$

namely the x-axis.

This algebraic set is irreducible, and prime ideals correspond to irreducible algebraic sets.

Another example is

$$(y - x^2).$$

Since

$$k[x, y]/(y - x^2) \cong k[x]$$

is again a domain, this ideal is prime as well. Geometrically, the parabola is irreducible.

Key principle. If I is prime, then $k[x, y]/I$ has no zero divisors. This algebraically reflects geometric irreducibility.

7. A non-prime ideal: reducible geometry

Take

$$I = (xy) \subseteq k[x, y].$$

Geometry. The equation

$$xy = 0$$

means

$$x = 0 \quad \text{or} \quad y = 0.$$

Thus

$$V(xy) = V(x) \cup V(y),$$

which is the union of the two coordinate axes.

Why (xy) is not prime. We have

$$xy \in (xy),$$

but

$$x \notin (xy), \qquad y \notin (xy).$$

So the defining property of prime ideals fails.

Quotient. Consider

$$k[x, y]/(xy).$$

In this quotient ring, the classes of x and y are both nonzero, but

$$\bar{x}\,\bar{y} = 0.$$

So the quotient ring has zero divisors.

Interpretation. This exactly reflects the fact that the variety $V(xy)$ has two irreducible components. Reducibility on the geometric side corresponds to zero divisors on the algebraic side.

8. Radical ideals: no hidden nilpotents

An ideal I is called radical if

$$f^n \in I \Longrightarrow f \in I \qquad \text{for all } f \in k[x,y],\ n \geq 1.$$

Equivalently,

$$I = \sqrt{I},$$

where

$$\sqrt{I} = \{f : f^n \in I \text{ for some } n \geq 1\}.$$

Example of a radical ideal. The ideal

$$(xy)$$

is radical in $k[x,y]$, because

$$(xy) = (x) \cap (y).$$

It defines the reduced union of the two coordinate axes.

Example of a non-radical ideal. Take

$$I = (x^2) \subseteq k[x].$$

This is not radical, since

$$x^2 \in (x^2), \qquad x \notin (x^2).$$

Its radical is

$$\sqrt{(x^2)} = (x).$$

Quotient ring. In

$$k[x]/(x^2),$$

the class of x is nonzero, but

$$x^2 = 0.$$

Thus the quotient contains a nonzero nilpotent element.

Geometric meaning. The ideals (x) and (x^2) define the same point set:

$$V(x) = V(x^2) = \{0\}.$$

But they do not define the same algebra. The ideal (x^2) carries extra infinitesimal structure. Radical ideals correspond to reduced geometry, while non-radical ideals produce nilpotent thickening.

9. Primary ideals: one prime with multiplicity

An ideal Q is primary if

$$ab \in Q,\ a \notin Q \Longrightarrow b^n \in Q \text{ for some } n \geq 1.$$

Example in one variable. The ideal

$$Q = (x^2) \subseteq k[x]$$

is primary. Its radical is

$$\sqrt{Q} = (x),$$

which is prime. So (x^2) is (x)-primary.

Geometric meaning. This describes the point $x = 0$, but with multiplicity or nilpotent thickening.

Example in two variables. Take

$$Q = (x, y^2) \subseteq k[x, y].$$

Its radical is

$$\sqrt{Q} = (x, y),$$

so Q is (x, y)-primary.

Its zero set is still just the origin:

$$V(x, y^2) = \{(0, 0)\}.$$

But in the quotient

$$k[x, y]/(x, y^2),$$

the class of y is nonzero and satisfies

$$\bar{y}^2 = 0.$$

Thus we have again a nonreduced structure concentrated at a single geometric point.

Interpretation. Primary ideals describe one geometric piece whose underlying support is irreducible, but which may carry multiplicity or nilpotent behavior.

10. Minimal primes and decomposition on a concrete example

Take again

$$I = (xy) \subseteq k[x, y].$$

Then

$$(xy) = (x) \cap (y).$$

The ideals (x) and (y) are prime, and they are the minimal prime ideals over (xy).

Geometry. These correspond to the two irreducible components:

- (x) corresponds to the y-axis,
- (y) corresponds to the x-axis.

Thus the decomposition of the ideal reflects the decomposition of the algebraic set into irreducible components.

Now compare with

$$I = (x^2y).$$

Then

$$\sqrt{(x^2y)} = (xy) = (x) \cap (y).$$

So the underlying set of points is the same as for (xy), namely the union of the two axes, but the algebraic structure is different: one component carries multiplicity.

11. Coordinate rings: several important examples

Example A: a line. Let

$$X = V(y) \subseteq k^2.$$

Then

$$k[X] = k[x, y]/(y) \cong k[x].$$

So polynomial functions on the line are just polynomials in one variable.

Example B: a parabola. Let

$$X = V(y - x^2).$$

Then

$$k[X] = k[x, y]/(y - x^2) \cong k[x].$$

So although the parabola sits in the plane, its coordinate ring behaves like a one-variable polynomial ring.

Example C: a cusp. Let

$$X = V(y^2 - x^3).$$

Then

$$k[X] = k[x, y]/(y^2 - x^3).$$

This ring is a domain, so the curve is irreducible. However, it is more subtle than the parabola. A parametrization is

$$x = t^2, \qquad y = t^3.$$

Hence

$$k[X] \cong k[t^2, t^3] \subseteq k[t].$$

This shows concretely that the coordinate ring is not simply a full polynomial ring $k[t]$, but rather a subring generated by t^2 and t^3. The singularity at the origin is encoded in this algebraic difference.

Example D: union of axes. Let

$$X = V(xy).$$

Then

$$k[X] = k[x, y]/(xy).$$

This ring has zero divisors, reflecting the fact that X is reducible.

Example E: a doubled point. Take the ideal $(x^2) \subseteq k[x]$. Then the coordinate ring is

$$k[x]/(x^2).$$

Its point set is just $\{0\}$, but it is not reduced. This is the simplest example of a nonreduced scheme-like object.

12. What elements of quotient rings actually look like

It is important to understand that an element of a quotient ring is a class of polynomials modulo the given relations.

Example 1. Let

$$R = k[x, y]/(y - x^2).$$

An element is a class

$$[f(x, y)].$$

But because $y = x^2$ in the quotient, every class can be simplified to a polynomial in x alone. For example:

$$[y] = [x^2],$$
$$[x^2 + y] = [2x^2],$$
$$[y^3 + x] = [x^6 + x].$$

So although the quotient was defined from a two-variable ring, its elements have a simple normal form.

Example 2. Let

$$R = k[x, y]/(xy).$$

Now the relation is

$$xy = 0.$$

But this does not allow us to eliminate one variable completely. Instead, we keep the rule that every product containing a factor xy is zero. Thus:

$$[x^2y] = [x(xy)] = 0,$$

$$[xy^7] = 0,$$

while

$$[x + y] \neq 0, \qquad [x] \neq 0, \qquad [y] \neq 0, \qquad [x][y] = 0.$$

So quotient rings store the relations exactly as algebraic rules among classes of polynomials.

13. Localization: concrete algebraic meaning

Localization means making certain polynomials invertible.

Example 1: invert x. Let

$$S = \{1, x, x^2, x^3, \dots\} \subseteq k[x, y].$$

Then

$$S^{-1}k[x, y] = k[x, x^{-1}, y].$$

So now x is invertible.

Geometric meaning. This corresponds to restricting the affine plane to the open subset

$$D(x) = \{(a, b) \in k^2 : a \neq 0\}.$$

Thus localizing at x means studying the part of the variety where $x \neq 0$.

Example 2: localize the parabola away from the origin. Start with

$$A = k[x, y]/(y - x^2) \cong k[x].$$

Localize at powers of x:

$$A_x \cong k[x, x^{-1}].$$

This studies the parabola on the open subset where $x \neq 0$, equivalently away from its origin.

Example 3: localization at a maximal ideal. Take

$$A = k[x, y], \qquad \mathfrak{m} = (x, y).$$

Then

$$A_{\mathfrak{m}}$$

consists of fractions

$$\frac{f}{g}$$

where $g \notin (x, y)$. This means $g(0, 0) \neq 0$, so denominators are exactly the polynomials nonvanishing at the origin.

Examples of elements:

$$\frac{x^2 + y}{1 + x + y}, \qquad \frac{x}{1 - y}, \qquad \frac{y^3}{2 + x^5}.$$

But

$$\frac{1}{x}$$

does not belong to $A_{\mathfrak{m}}$, because $x \in (x, y)$, so x vanishes at the origin.

Meaning. The local ring $A_{\mathfrak{m}}$ captures functions regular near $(0, 0)$. It is the algebraic ring of local behavior near that point.

Example 4: local ring of the parabola at its origin. Let

$$A = k[x, y]/(y - x^2), \qquad \mathfrak{m} = (x, y)/(y - x^2).$$

Then

$$A_{\mathfrak{m}}$$

is the local ring of the parabola at the origin. Since $A \cong k[x]$, this is just

$$k[x]_{(x)}.$$

Its elements are fractions

$$\frac{f(x)}{g(x)} \qquad \text{with } g(0) \neq 0.$$

Thus this ring sees only the behavior of the parabola near the point $x = 0$.

Example 5: localization at a prime ideal. Take

$$A = k[x, y], \qquad \mathfrak{p} = (y).$$

Then

$$A_{\mathfrak{p}}$$

has denominators all polynomials not divisible by y. In particular, x becomes invertible, because $x \notin (y)$, but y does not become invertible.

Geometric meaning. This is not localization at a point. Rather, it is localization near the irreducible subvariety $V(y)$, namely the x-axis. Thus:

- localization at a maximal ideal means near a point,
- localization at a prime ideal means near an irreducible subvariety.

14. Quotients and localization together

These operations combine naturally and are fundamental in local algebraic geometry.

Take

$$A = k[x, y], \qquad I = (y - x^2).$$

First form the quotient

$$A/I = k[x, y]/(y - x^2).$$

Then localize at the image of (x, y):

$$(A/I)_{(x,y)/(y-x^2)}.$$

This is the local coordinate ring of the parabola at the origin.

Equivalently, one may first localize A at (x, y), then quotient by the extended ideal:

$$A_{(x,y)}/IA_{(x,y)}.$$

These two rings are naturally isomorphic.

Interpretation.

- Quotienting imposes the equations of the variety.
- Localization zooms in near a chosen point or subvariety.

Together they give the algebra of local geometry.

15. A comparison table of key examples

Ideal	Quotient ring	Geometry	Type
$(x-a, y-b)$	$k[x,y]/(x-a,y-b) \cong k$	one point	maximal, prime, radical
(y)	$k[x,y]/(y) \cong k[x]$	line	prime, radical, not maximal
$(y-x^2)$	$k[x,y]/(y-x^2) \cong k[x]$	parabola	prime, radical
(xy)	$k[x,y]/(xy)$	two axes crossing	radical, not prime
$(x^2) \subseteq k[x]$	$k[x]/(x^2)$	doubled point	primary, not radical, not prime
(x, y^2)	$k[x,y]/(x,y^2)$	thickened origin	(x,y)-primary

16. Deep conceptual summary

It is useful to collect the meaning of the main notions in one place.

Ideal. An ideal records polynomial relations.

Quotient ring. The quotient by an ideal is the algebra where those relations are literally true.

Coordinate ring. A quotient ring interpreted as the ring of polynomial functions on the corresponding algebraic set.

Prime ideal. Corresponds to an irreducible algebraic set; the quotient is a domain.

Maximal ideal. Corresponds to a point; the quotient is a field.

Radical ideal. Corresponds to reduced geometry, with no nilpotent structure.

Primary ideal. Corresponds to one geometric component with multiplicity or infinitesimal thickening.

Localization. Means looking only near a point, near a component, or on an open subset where certain functions do not vanish.

17. One full worked chain: the example (xy)

Let
$$I = (xy) \subseteq k[x, y].$$

Step 1: the ideal. It encodes the equation
$$xy = 0.$$

Step 2: the zero set. The variety is
$$V(I) = V(x) \cup V(y),$$
the union of the two coordinate axes.

Step 3: the quotient ring. Set
$$A = k[x, y]/(xy).$$
In A, one has
$$\bar{x}\bar{y} = 0.$$
So A has zero divisors.

Step 4: the minimal primes. The minimal prime ideals over (xy) are
$$(x), \qquad (y).$$
These correspond to the two irreducible components.

Step 5: localization at the origin. If we localize at the maximal ideal
$$(x, y)/(xy),$$
we obtain
$$A_{(x,y)/(xy)}.$$
This ring describes the crossing only near the origin.

Step 6: localization near one component. If instead we localize at the prime ideal
$$(x)/(xy),$$
we focus on the component $x = 0$, near its generic behavior.

Conclusion. This one example shows simultaneously:

- an ideal,
- its zero set,
- a quotient ring,
- reducibility,
- minimal primes,
- localization near a point,
- localization near a component.

18. Best examples to remember

For intuition, the following examples are the most useful basic models:

$$k[x,y]/(y-x^2) \qquad \text{parabola,}$$

$$k[x,y]/(xy) \qquad \text{two axes crossing,}$$

$$k[x]/(x^2) \qquad \text{nonreduced doubled point,}$$

$$k[x,y]/(x,y) \cong k \qquad \text{single point,}$$

$$k[x,y]_{(x,y)} \qquad \text{functions regular near the origin,}$$

$$\big(k[x,y]/(y-x^2)\big)_{(x,y)/(y-x^2)} \qquad \text{local ring of the parabola at the origin.}$$

These examples cover most of the core intuition behind:

- ideals,
- quotient rings,
- coordinate rings,
- radical and primary structure,
- local behavior via localization.

19. Final synthesis

The central chain of ideas is:

$$\text{polynomial ring} \longrightarrow \text{ideal of relations} \longrightarrow \text{quotient ring} \longrightarrow \text{coordinate ring of a geometric object}$$

Concretely:

- $k[x,y]$ is the ambient ring of polynomial expressions,
- an ideal such as $(y-x^2)$ or (xy) imposes algebraic equations,
- the quotient ring encodes those equations as true identities,
- the coordinate ring describes polynomial functions on the resulting algebraic set,
- localization refines the picture by zooming in on an open set, a point, or an irreducible piece.

In this way, algebra and geometry mirror each other:

- points correspond to maximal ideals,
- irreducible sets correspond to prime ideals,
- reducedness corresponds to radical ideals,
- multiplicity and infinitesimal thickening correspond to primary and non-radical structure,
- local behavior corresponds to localization.

That is the deep reason polynomial rings, ideals, quotient rings, coordinate rings, and localization fit together so naturally.

Problem 8 — Nagata's localization example

Exact statement. Let k be a field and let

$$A = k[x_1, x_2, x_3, \dots]$$

be the polynomial ring in countably many variables. Let

$$m_1, m_2, \dots$$

be an increasing sequence of positive integers such that

$$m_{i+1} - m_i > m_i - m_{i-1} \qquad \text{for all } i > 1.$$

Let

$$\mathfrak{p}_i = (x_{m_i+1}, \dots, x_{m_{i+1}})$$

and let

$$S = A \setminus \bigcup_{i=1}^{\infty} \mathfrak{p}_i.$$

Show that:

(a) S is multiplicatively closed.

(b) The maximal ideals of $S^{-1}A$ are the localizations $S^{-1}\mathfrak{p}_i$.

(c) Using Q7, show that $S^{-1}A$ is Noetherian.

(d) The height of $S^{-1}\mathfrak{p}_i$ is $m_{i+1} - m_i$, and hence

$$\dim S^{-1}A = \infty.$$

Solution. Set

$$d_i := m_{i+1} - m_i.$$

Then the hypothesis says

$$d_1 < d_2 < d_3 < \cdots,$$

so in particular the d_i are unbounded.

Also, each $\mathfrak{p}_i$ is a prime ideal, because

$$A/\mathfrak{p}_i \cong k[x_1, \dots, x_{m_i}, x_{m_{i+1}+1}, x_{m_{i+1}+2}, \dots]$$

is a domain.

(a) S is multiplicatively closed.
Take $f, g \in S$. Suppose $fg \notin S$. Then

$$fg \in \bigcup_{i=1}^{\infty} \mathfrak{p}_i,$$

so for some i,

$$fg \in \mathfrak{p}_i.$$

Since $\mathfrak{p}_i$ is prime, it follows that

$$f \in \mathfrak{p}_i \quad \text{or} \quad g \in \mathfrak{p}_i,$$

contrary to $f, g \in S$. Hence $fg \in S$, so S is multiplicatively closed.

(b) The maximal ideals of $S^{-1}A$ are exactly $S^{-1}\mathfrak{p}_i$.

Primes of $S^{-1}A$ correspond to primes $\mathfrak{q} \subseteq A$ such that

$$\mathfrak{q} \cap S = \varnothing.$$

Thus it is enough to show that if $\mathfrak{q}$ is prime and $\mathfrak{q} \cap S = \varnothing$, then

$$\mathfrak{q} \subseteq \mathfrak{p}_i$$

for some i.

Now

$$\mathfrak{q} \cap S = \varnothing \quad \Longleftrightarrow \quad \mathfrak{q} \subseteq \bigcup_{i=1}^{\infty} \mathfrak{p}_i.$$

If $\mathfrak{q} = (0)$, this is trivial. So assume $\mathfrak{q} \neq (0)$, and choose $0 \neq f \in \mathfrak{q}$.

Since f is a polynomial in only finitely many variables, there exists N_0 such that

$$f \in A_{N_0} := k[x_1, \dots, x_{m_{N_0+1}}].$$

For $N \geq N_0$, let

$$\mathfrak{q}_N := \mathfrak{q} \cap A_N.$$

Then $\mathfrak{q}_N$ is a nonzero prime ideal of the Noetherian ring A_N.

For $i > N$, one has

$$\mathfrak{p}_i \cap A_N = (0),$$

because the generators of $\mathfrak{p}_i$ involve only variables with indices $> m_{N+1}$. Hence

$$\mathfrak{q}_N \subseteq \bigcup_{i=1}^{N} (\mathfrak{p}_i \cap A_N).$$

By finite prime avoidance,

$$\mathfrak{q}_N \subseteq \mathfrak{p}_{i_N} \cap A_N$$

for some $i_N \in \{1, \dots, N\}$.

Now

$$\mathfrak{q}_{N_0} = \mathfrak{q}_N \cap A_{N_0} \subseteq (\mathfrak{p}_{i_N} \cap A_N) \cap A_{N_0}.$$

If $i_N > N_0$, then $\mathfrak{p}_{i_N} \cap A_{N_0} = (0)$, so $\mathfrak{q}_{N_0} = 0$, contradiction. Therefore, for every $N \geq N_0$,

$$i_N \in \{1, \dots, N_0\}.$$

So only finitely many indices occur. Hence some fixed $i \in \{1, \dots, N_0\}$ occurs for infinitely many N.

Take any $g \in \mathfrak{q}$. Then $g \in A_M$ for some M. Choose $N \geq \max\{M, N_0\}$ such that $i_N = i$. Then

$$g \in \mathfrak{q}_N \subseteq \mathfrak{p}_i \cap A_N,$$

so $g \in \mathfrak{p}_i$. Therefore

$$\mathfrak{q} \subseteq \mathfrak{p}_i.$$

Thus every prime disjoint from S is contained in some $\mathfrak{p}_i$, and so the maximal such primes are exactly the $\mathfrak{p}_i$. Therefore the maximal ideals of $S^{-1}A$ are exactly

$$S^{-1}\mathfrak{p}_i.$$

(c) $S^{-1}A$ **is Noetherian.**

By part (b), the maximal ideals of $S^{-1}A$ are the ideals $S^{-1}\mathfrak{p}_i$. Localizing further at one of them gives

$$(S^{-1}A)_{S^{-1}\mathfrak{p}_i} \cong A_{\mathfrak{p}_i}.$$

Let

$$B_i = k\big[x_j \mid j \notin \{m_i + 1, \dots, m_{i+1}\}\big].$$

Then B_i is a domain, and every nonzero element of B_i lies outside $\mathfrak{p}_i$, hence becomes a unit in $A_{\mathfrak{p}_i}$. If

$$K_i = \mathrm{Frac}(B_i),$$

then

$$A_{\mathfrak{p}_i} \cong K_i[y_1, \dots, y_{d_i}]_{(y_1,\dots,y_{d_i})},$$

where

$$y_r = x_{m_i+r}, \qquad d_i = m_{i+1} - m_i.$$

This is a localization of a polynomial ring in finitely many variables over a field, hence it is Noetherian.

Therefore every localization of $S^{-1}A$ at a maximal ideal is Noetherian. By the criterion from Q7, it follows that

$$S^{-1}A$$

is Noetherian.

(d) $\mathrm{ht}(S^{-1}\mathfrak{p}_i) = m_{i+1} - m_i$**, hence** $\dim S^{-1}A = \infty$**.**

Again,

$$(S^{-1}A)_{S^{-1}\mathfrak{p}_i} \cong A_{\mathfrak{p}_i} \cong K_i[y_1, \dots, y_{d_i}]_{(y_1,\dots,y_{d_i})}.$$

The local ring on the right has dimension d_i. Therefore

$$\mathrm{ht}(S^{-1}\mathfrak{p}_i) = d_i = m_{i+1} - m_i.$$

Since

$$d_1 < d_2 < d_3 < \cdots,$$

the heights of these maximal ideals are unbounded. Hence

$$\dim S^{-1}A = \infty.$$

So $S^{-1}A$ is a Noetherian ring of infinite Krull dimension.

Problem 9 — Dimension bounds for a polynomial ring

Exact statement. Let A be a ring (not necessarily Noetherian). Show that

$$1 + \dim A \leq \dim A[x] \leq 1 + 2 \dim A.$$

Solution.
We prove the two inequalities separately.

Lower bound: $1 + \dim A \leq \dim A[x]$**.**
Take a strict chain of prime ideals in A:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

Then in $A[x]$ we obtain

$$\mathfrak{p}_0 A[x] \subsetneq \mathfrak{p}_1 A[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n A[x] \subsetneq \mathfrak{p}_n A[x] + (x).$$

The last ideal is prime because

$$A[x]/(\mathfrak{p}_n A[x] + (x)) \cong (A/\mathfrak{p}_n)[x]/(x) \cong A/\mathfrak{p}_n,$$

which is a domain. Hence every chain of length n in A yields a chain of length $n+1$ in $A[x]$. Therefore

$$\dim A[x] \geq \dim A + 1.$$

Upper bound: $\dim A[x] \leq 1 + 2 \dim A$**.**
If $\dim A = \infty$, there is nothing to prove. So assume $\dim A < \infty$.
Take a chain of prime ideals in $A[x]$:

$$Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_m.$$

Let

$$\mathfrak{p}_j := Q_j \cap A.$$

Then

$$\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_m.$$

Delete repetitions and write the distinct contractions as

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_r.$$

Then

$$r \leq \dim A.$$

Now the original chain splits into consecutive blocks with constant contraction. Fix one such contraction $\mathfrak{q}$. The corresponding block lies in the fiber of

$$f^* : \operatorname{Spec}(A[x]) \to \operatorname{Spec}(A)$$

over $\mathfrak{q}$, where $f : A \to A[x]$ is the natural inclusion.
By the standard description of fibers,

$$(f^*)^{-1}(\mathfrak{q}) \cong \operatorname{Spec}(k(\mathfrak{q})[x]).$$

Since $k(\mathfrak{q})$ is a field,

$$\dim k(\mathfrak{q})[x] = 1.$$

Thus any chain of primes in a single fiber has length at most 1, so each block contains at most two prime ideals.

There are $r+1$ distinct contractions, hence the total number of primes in the chain is at most

$$2(r+1).$$

Therefore

$$m \le 2(r+1) - 1 = 2r + 1 \le 2\dim A + 1.$$

Taking the supremum over all chains gives

$$\dim A[x] \le 1 + 2\dim A.$$

Combining the two inequalities, we obtain

$$1 + \dim A \le \dim A[x] \le 1 + 2\dim A.$$

Theory for Problems 8–9

Problem 8 — Theory behind Nagata's localization example

Theory. We begin with

$$A = k[x_1, x_2, x_3, \dots],$$

the polynomial ring in countably many variables. This ring is not Noetherian, since there is an infinite strictly increasing chain of ideals

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots .$$

Nagata's idea is to localize A at a carefully chosen multiplicative set S so that the resulting ring $S^{-1}A$ becomes Noetherian, while still having infinite Krull dimension. This is one of the standard examples showing that a Noetherian ring need not have finite dimension.

We choose integers

$$m_1 < m_2 < \cdots$$

such that

$$d_i := m_{i+1} - m_i$$

is strictly increasing:

$$d_1 < d_2 < d_3 < \cdots .$$

Then define prime ideals

$$\mathfrak{p}_i = (x_{m_i+1}, \dots, x_{m_{i+1}}).$$

Each $\mathfrak{p}_i$ is prime because

$$A/\mathfrak{p}_i \cong k[\text{all variables except } x_{m_i+1}, \dots, x_{m_{i+1}}],$$

which is again a polynomial ring over a field, hence a domain.

Now set
$$S = A \setminus \bigcup_{i \geq 1} \mathfrak{p}_i.$$
The complement of a union of prime ideals is multiplicatively closed: if $ab \in \mathfrak{p}_i$ for some i, then since $\mathfrak{p}_i$ is prime, either $a \in \mathfrak{p}_i$ or $b \in \mathfrak{p}_i$. Therefore if $a, b \notin \bigcup_i \mathfrak{p}_i$, then also $ab \notin \bigcup_i \mathfrak{p}_i$.

The difficult point is that the union
$$\bigcup_{i \geq 1} \mathfrak{p}_i$$
is infinite, so ordinary finite prime avoidance does not apply directly. For a finite union of primes, if a prime $\mathfrak{q}$ is contained in the union, then $\mathfrak{q}$ must lie inside one of them. For an infinite union, this need not be true in general.

The special feature of the present situation is that every polynomial involves only finitely many variables. Thus if a prime $\mathfrak{q} \subseteq A$ satisfies
$$\mathfrak{q} \cap S = \varnothing,$$
equivalently
$$\mathfrak{q} \subseteq \bigcup_{i \geq 1} \mathfrak{p}_i,$$
then by intersecting $\mathfrak{q}$ with a sufficiently large finite polynomial subring
$$A_N = k[x_1, \dots, x_{m_{N+1}}],$$
one reduces to a finite-stage problem, where only finitely many of the $\mathfrak{p}_i$ are visible. Inside A_N, finite prime avoidance applies. This is the key idea: although the ambient ring has infinitely many variables, each individual polynomial lives in a finite-dimensional stage.

Primes of the localization $S^{-1}A$ correspond to primes $\mathfrak{q} \subseteq A$ with $\mathfrak{q} \cap S = \varnothing$. The finite-support argument shows that every such prime must actually satisfy
$$\mathfrak{q} \subseteq \mathfrak{p}_i$$
for some i. Hence the maximal ideals of $S^{-1}A$ are exactly
$$S^{-1}\mathfrak{p}_i.$$

Now localize further at one of these maximal ideals:
$$(S^{-1}A)_{S^{-1}\mathfrak{p}_i} \cong A_{\mathfrak{p}_i}.$$
Inside $A_{\mathfrak{p}_i}$, every variable outside the block
$$x_{m_i+1}, \dots, x_{m_{i+1}}$$
becomes a unit, because it does not lie in $\mathfrak{p}_i$. Thus all those outside variables are absorbed into the coefficient field. If
$$K_i = \mathrm{Frac}\big(k[\text{all variables outside the } i\text{-th block}]\big),$$

then
$$A_{\mathfrak{p}_i} \cong K_i[y_1, \ldots, y_{d_i}]_{(y_1,\ldots,y_{d_i})}, \qquad d_i = m_{i+1} - m_i.$$
So each maximal localization is a standard regular local ring in finitely many variables over a field. In particular, it is Noetherian and has dimension d_i.

Therefore the heights of the maximal ideals are
$$\mathrm{ht}(S^{-1}\mathfrak{p}_i) = d_i = m_{i+1} - m_i.$$
Since the numbers d_i are strictly increasing and unbounded, the maximal ideals have arbitrarily large height, and so
$$\dim S^{-1}A = \infty.$$

Thus Nagata's example shows:

- localization can turn a non-Noetherian ring into a Noetherian one;
- Noetherianity does not imply finite Krull dimension;
- local analysis at maximal ideals can control a globally complicated ring.

Problem 9 — Theory behind the dimension bounds for $A[x]$

Theory. The problem asks how the Krull dimension changes when one adjoins one polynomial variable:
$$A \longmapsto A[x].$$
For Noetherian rings, the classical formula is
$$\dim A[x] = \dim A + 1.$$
Without the Noetherian hypothesis, equality may fail, but one still has the general bounds
$$1 + \dim A \le \dim A[x] \le 1 + 2\dim A.$$
This is part of the dimension theory developed by Seidenberg.

The main structural tool is the natural ring map
$$A \hookrightarrow A[x],$$
which induces a map on spectra
$$f^* : \mathrm{Spec}(A[x]) \to \mathrm{Spec}(A), \qquad Q \mapsto Q \cap A.$$
Thus every prime of $A[x]$ lies over a prime of A. One studies chains in $\mathrm{Spec}(A[x])$ by looking at how they project to $\mathrm{Spec}(A)$.

Fix $\mathfrak{p} \in \mathrm{Spec}(A)$. The fiber over $\mathfrak{p}$ is
$$(f^*)^{-1}(\mathfrak{p}).$$
A standard calculation identifies this fiber with
$$\mathrm{Spec}(k(\mathfrak{p})[x]),$$

where

$$k(\mathfrak{p}) = \mathrm{Frac}(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}})$$

is the residue field at $\mathfrak{p}$. Since $k(\mathfrak{p})[x]$ is a polynomial ring in one variable over a field, it has Krull dimension 1. Therefore every fiber has dimension 1.

This gives the geometric picture:

- the spectrum of $A[x]$ sits over the spectrum of A;
- the extra variable x contributes a vertical fiber of dimension 1 over each point of the base.

The lower bound

$$\dim A[x] \geq \dim A + 1$$

is easy. If

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

is a chain in A, then

$$\mathfrak{p}_0 A[x] \subsetneq \mathfrak{p}_1 A[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n A[x] \subsetneq \mathfrak{p}_n A[x] + (x)$$

is a chain in $A[x]$. The last ideal is prime because

$$A[x]/(\mathfrak{p}_n A[x] + (x)) \cong (A/\mathfrak{p}_n)[x]/(x) \cong A/\mathfrak{p}_n,$$

which is a domain.

For the upper bound, take a chain

$$Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_m$$

in $A[x]$, and contract to A:

$$\mathfrak{p}_i = Q_i \cap A.$$

Then

$$\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_m.$$

If we remove repetitions, we obtain a strictly increasing chain

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_r$$

in A, so $r \leq \dim A$.

Now split the original chain into consecutive blocks where the contraction is constant. Each such block lies entirely inside one fiber of f^*. But every fiber is isomorphic to $\mathrm{Spec}(k(\mathfrak{q})[x])$, which has dimension 1. Therefore each block can contain at most two distinct primes. Since there are $r+1$ distinct contractions, the total number of primes in the chain is at most

$$2(r+1),$$

so the length of the chain is at most

$$2(r+1) - 1 = 2r + 1 \leq 2\dim A + 1.$$

Hence

$$\dim A[x] \leq 1 + 2\dim A.$$

The conceptual reason for the upper bound is that a chain in $A[x]$ can move in two ways:

- horizontally, by changing its contraction in $\mathrm{Spec}(A)$;
- vertically, by moving inside a fiber of dimension 1.

In the Noetherian case, one has stronger control and gets the exact equality

$$\dim A[x] = \dim A + 1.$$

Without Noetherianity, several primes can occur over the same contraction in a less rigid way, and only the weaker Seidenberg bound remains.

Problems 8 and 9 complement each other well:

- Problem 8 shows that a Noetherian ring can have infinite dimension.
- Problem 9 shows that polynomial extension behaves more subtly once one leaves the Noetherian world.

Together they illustrate the central role of localization, fibers of Spec, and chain arguments in commutative algebra.

Theory add-on: spectra, kinds of points, and deep examples

1. What is $\mathrm{Spec}(A)$?

For a commutative ring A, the spectrum

$$\mathrm{Spec}(A)$$

is the set of all prime ideals of A. Thus a point of $\mathrm{Spec}(A)$ is a prime ideal

$$\mathfrak{p} \subseteq A.$$

This is the basic bridge between algebra and geometry: ideals, quotients, dimension, and localization are encoded as geometric features of the spectrum.

2. Zariski closed sets

For an ideal $I \subseteq A$, define

$$V(I) := \{\mathfrak{p} \in \mathrm{Spec}(A) \mid I \subseteq \mathfrak{p}\}.$$

These are the closed sets of the Zariski topology.

Important special case:

$$V(f) := V((f)) = \{\mathfrak{p} \mid f \in \mathfrak{p}\}.$$

Also,

$$V(I) = V(\sqrt{I}),$$

so closed sets depend only on radicals.

3. Basic open sets and localization

For $f \in A$, define

$$D(f) := \mathrm{Spec}(A) \setminus V(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \notin \mathfrak{p}\}.$$

This corresponds to localization:

$$D(f) \quad \leftrightarrow \quad A_f.$$

More generally, for a multiplicative set $S \subseteq A$,

$$\mathrm{Spec}(S^{-1}A) \cong \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \varnothing\}.$$

This is exactly the mechanism behind Problem 8.

4. Closure of a point and specialization

For a point $\mathfrak{p} \in \mathrm{Spec}(A)$, its closure is

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}) = \{\mathfrak{q} \in \mathrm{Spec}(A) \mid \mathfrak{p} \subseteq \mathfrak{q}\}.$$

Thus inclusion of prime ideals is the geometric specialization relation:

$$\mathfrak{p} \subseteq \mathfrak{q} \quad \Longleftrightarrow \quad \mathfrak{q} \text{ is a specialization of } \mathfrak{p}.$$

5. Kinds of points

Closed points. A point $\mathfrak{p}$ is closed iff $\mathfrak{p}$ is maximal.

Generic points. A point $\mathfrak{p}$ is generic for an irreducible closed subset Y if

$$\overline{\{\mathfrak{p}\}} = Y.$$

Every irreducible closed subset of $\mathrm{Spec}(A)$ has the form $V(\mathfrak{p})$ for a prime ideal $\mathfrak{p}$.

Minimal points. Minimal primes are the generic points of the irreducible components of $\mathrm{Spec}(A)$.

Height-r points. A prime ideal $\mathfrak{p}$ has height

$$\mathrm{ht}(\mathfrak{p}) = \sup\{n \mid \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}\}.$$

This is the codimension of the point.

6. Kinds of spectra

Spectrum of a field. If $A = k$ is a field, then

$$\mathrm{Spec}(k) = \{(0)\}.$$

Spectrum of a domain. If A is a domain, then (0) is prime, and in fact

$$A \text{ is a domain} \iff \mathrm{Spec}(A) \text{ is irreducible.}$$

Spectrum of a local ring. If A is local with maximal ideal m, then $\mathrm{Spec}(A)$ has a unique closed point, namely m.

Spectrum of a product. If $A = A_1 \times A_2$, then

$$\mathrm{Spec}(A) \cong \mathrm{Spec}(A_1) \sqcup \mathrm{Spec}(A_2).$$

7. Deep example: $\mathrm{Spec}(k[x])$

Let k be a field. Then the prime ideals of $k[x]$ are:

$$(0), \qquad (f) \text{ for irreducible } f.$$

Hence

$$\mathrm{Spec}(k[x]) = \{(0)\} \cup \{(f) : f \text{ irreducible}\}.$$

If k is algebraically closed, the irreducible polynomials are linear, so

$$\mathrm{Spec}(k[x]) = \{(0)\} \cup \{(x - a) : a \in k\}.$$

Here (0) is the generic point of the whole affine line, and the ideals $(x - a)$ are the closed points.

8. Deep example: $\mathrm{Spec}(\mathbb{Z})$

The prime ideals of $\mathbb{Z}$ are

$$(0), \qquad (p) \text{ for prime numbers } p.$$

So

$$\mathrm{Spec}(\mathbb{Z}) = \{(0)\} \cup \{(p) : p \text{ prime}\}.$$

The point (0) is generic, while the ideals (p) are the closed points.

9. Deep example: $\mathrm{Spec}(k[x, y])$

Assume k algebraically closed. Then important primes are:

- (0), the generic point of the whole plane;
- (f) for irreducible f, the generic points of irreducible curves;
- $(x - a, y - b)$, the closed points.

A typical chain is

$$(0) \subsetneq (f) \subsetneq (x - a, y - b),$$

so

$$\dim k[x, y] = 2.$$

10. Deep example: reducible spectrum

Let

$$A = k[x, y]/(xy).$$

Since

$$(xy) = (x) \cap (y),$$

the minimal primes are (x) and (y). Thus $\mathrm{Spec}(A)$ has two irreducible components, corresponding to the two coordinate axes.

11. Deep example: local spectrum

Take

$$A = k[x, y]_{(x,y)}.$$

Then $\operatorname{Spec}(A)$ has unique closed point $(x, y)A$, but also nonclosed points such as

$$(0), \quad (x)A, \quad (y)A, \quad (x - y)A.$$

Localization geometrically means zooming into a neighborhood of one point.

12. Deep example: spectra in Problem 8

In Problem 8 one considers

$$A = k[x_1, x_2, \dots]$$

and

$$S = A \setminus \bigcup_i \mathfrak{p}_i.$$

Then

$$\operatorname{Spec}(S^{-1}A) \cong \{\mathfrak{q} \in \operatorname{Spec}(A) \mid \mathfrak{q} \cap S = \varnothing\}.$$

So one keeps exactly the primes disjoint from S. The problem shows that these are controlled by the primes $\mathfrak{p}_i$, and the maximal ideals become $S^{-1}\mathfrak{p}_i$. Their heights are unbounded, so the localized ring is Noetherian but infinite-dimensional.

13. Deep example: fibers in Problem 9

The inclusion

$$A \hookrightarrow A[x]$$

induces

$$\pi : \operatorname{Spec}(A[x]) \to \operatorname{Spec}(A).$$

For $\mathfrak{p} \in \operatorname{Spec}(A)$, the fiber is

$$\pi^{-1}(\mathfrak{p}) \cong \operatorname{Spec}(k(\mathfrak{p})[x]),$$

where

$$k(\mathfrak{p}) = \operatorname{Frac}(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}).$$

Since $k(\mathfrak{p})[x]$ has dimension 1, each fiber contributes at most one vertical step in a chain. This is the key geometric reason behind the estimate

$$1 + \dim A \le \dim A[x] \le 1 + 2 \dim A.$$

14. Dictionary

Algebra	Geometry
prime ideal $\mathfrak{p}$	point of $\operatorname{Spec}(A)$
maximal ideal	closed point
minimal prime	generic point of an irreducible component
$\mathfrak{p} \subseteq \mathfrak{q}$	$\mathfrak{q}$ is a specialization of $\mathfrak{p}$
A/I	closed subset $V(I)$
A_f	basic open subset $D(f)$
$S^{-1}A$	subspace of primes disjoint from S
$\dim A$	maximal length of chains of prime ideals

Theory add-on: maps of spectra, contraction, fibers, and geometric meaning

1. Ring maps induce maps of spectra

If

$$\varphi : A \to B$$

is a ring homomorphism, then every prime ideal $\mathfrak{q} \subseteq B$ determines a prime ideal

$$\varphi^{-1}(\mathfrak{q}) \subseteq A.$$

Hence there is an induced map

$$\varphi^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A), \qquad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}).$$

The direction reverses:

$$A \to B \qquad \Longrightarrow \qquad \mathrm{Spec}(B) \to \mathrm{Spec}(A).$$

2. Contraction and extension

If $\mathfrak{q} \subseteq B$ is prime, then $\varphi^{-1}(\mathfrak{q})$ is prime. This is because $A/\varphi^{-1}(\mathfrak{q})$ embeds into $B/\mathfrak{q}$, and $B/\mathfrak{q}$ is a domain.

If $I \subseteq A$ is an ideal, its extension to B is

$$IB := \varphi(I)B.$$

Unlike contraction, extension of a prime ideal need not remain prime.

3. Continuity

The induced map φ^* is continuous for the Zariski topology. Indeed, for any ideal $I \subseteq A$,

$$(\varphi^*)^{-1}(V(I)) = V(IB).$$

4. Quotients give closed immersions

For the quotient map

$$A \to A/I,$$

primes of A/I correspond exactly to primes of A containing I. Hence

$$\mathrm{Spec}(A/I) \cong V(I) \subseteq \mathrm{Spec}(A).$$

Thus quotient rings correspond geometrically to closed subsets.

Example:

$$A = k[x, y], \qquad I = (y - x^2).$$

Then

$$\mathrm{Spec}(A/I)$$

is the parabola $V(y - x^2)$ inside $\mathrm{Spec}(k[x, y])$.

5. Localization gives open immersions

Let $S \subseteq A$ be multiplicatively closed. Then

$$A \to S^{-1}A$$

induces

$$\operatorname{Spec}(S^{-1}A) \to \operatorname{Spec}(A),$$

whose image is

$$\{\mathfrak{p} \in \operatorname{Spec}(A) \mid \mathfrak{p} \cap S = \varnothing\}.$$

If $S = \{1, f, f^2, \dots\}$, then

$$\operatorname{Spec}(A_f) \cong D(f).$$

So localization corresponds to restricting to an open subset.

6. Localization at a prime

For a prime $\mathfrak{p} \subseteq A$,

$$A \to A_{\mathfrak{p}}$$

induces

$$\operatorname{Spec}(A_{\mathfrak{p}}) \cong \{\mathfrak{q} \in \operatorname{Spec}(A) \mid \mathfrak{q} \subseteq \mathfrak{p}\}.$$

So localization at $\mathfrak{p}$ keeps exactly the points specializing to $\mathfrak{p}$.

7. Polynomial extension

The inclusion

$$A \hookrightarrow A[x]$$

induces

$$\pi : \operatorname{Spec}(A[x]) \to \operatorname{Spec}(A).$$

A prime $Q \subseteq A[x]$ maps to $Q \cap A$. Geometrically, $\operatorname{Spec}(A[x])$ behaves like an affine line over $\operatorname{Spec}(A)$.

8. Fibers of $\operatorname{Spec}(A[x]) \to \operatorname{Spec}(A)$

For $\mathfrak{p} \in \operatorname{Spec}(A)$, the fiber is

$$\pi^{-1}(\mathfrak{p}) = \{Q \in \operatorname{Spec}(A[x]) \mid Q \cap A = \mathfrak{p}\}.$$

There is a canonical identification

$$\pi^{-1}(\mathfrak{p}) \cong \operatorname{Spec}(k(\mathfrak{p})[x]),$$

where

$$k(\mathfrak{p}) = \operatorname{Frac}(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}})$$

is the residue field. Since $k(\mathfrak{p})[x]$ has dimension 1, each fiber has dimension 1.

9. Surjectivity of polynomial projection

The map

$$\operatorname{Spec}(A[x]) \to \operatorname{Spec}(A)$$

is surjective: for every prime $\mathfrak{p} \subseteq A$, the ideal $\mathfrak{p}A[x]$ is prime and contracts to $\mathfrak{p}$.

10. Dominant maps

A map

$$f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$$

is called dominant if its image is dense in $\operatorname{Spec}(A)$. For $\varphi : A \to B$,

$$\varphi^* \text{ is dominant} \iff \ker(\varphi) \subseteq \sqrt{(0)}.$$

If $A \subseteq B$ are domains, then $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is dominant.

11. Integral extensions

If $A \to B$ is integral, then:

- every prime of A has a prime of B lying over it;
- chains can be lifted upward (going up);
- the induced map on spectra is surjective.

Example:

$$k[t^2, t^3] \subseteq k[t]$$

is integral and gives the normalization map of the cusp.

12. Residue fields

If $\mathfrak{q} \in \operatorname{Spec}(B)$ lies over $\mathfrak{p} = \mathfrak{q} \cap A$, then there is an induced field extension

$$k(\mathfrak{p}) \to k(\mathfrak{q}).$$

Thus a point upstairs may carry a larger residue field than the point downstairs.

13. Deep example: quotient map

For

$$k[x, y] \to k[x, y]/(xy),$$

one gets

$$\operatorname{Spec}(k[x, y]/(xy)) \cong V(xy) \subseteq \operatorname{Spec}(k[x, y]),$$

the union of the two coordinate axes.

14. Deep example: localization map

For
$$k[x,y] \to k[x,y]_x,$$
one gets
$$\mathrm{Spec}(k[x,y]_x) \cong D(x),$$
the part of the plane where $x \neq 0$.

15. Deep example: projection of the plane

For
$$k[y] \hookrightarrow k[x,y],$$
the induced map
$$\mathrm{Spec}(k[x,y]) \to \mathrm{Spec}(k[y])$$
is projection of the affine plane onto the y-line. The fiber over $(y-b)$ is
$$\mathrm{Spec}(k[x,y]/(y-b)) \cong \mathrm{Spec}(k[x]).$$

16. Deep example: arithmetic fibers

For
$$\mathbb{Z} \hookrightarrow \mathbb{Z}[x],$$
the fiber over (p) is
$$\mathrm{Spec}(\mathbb{F}_p[x]),$$
while the fiber over (0) is
$$\mathrm{Spec}(\mathbb{Q}[x]).$$
So the total space mixes all characteristics.

17. Connection with Problem 8

In Problem 8 one considers
$$A = k[x_1, x_2, \ldots], \qquad S = A \setminus \bigcup_i \mathfrak{p}_i.$$
Then
$$\mathrm{Spec}(S^{-1}A) \to \mathrm{Spec}(A)$$
identifies $\mathrm{Spec}(S^{-1}A)$ with the subspace of primes disjoint from S. The problem shows that the maximal points of this subspace are exactly the $S^{-1}\mathfrak{p}_i$.

18. Connection with Problem 9

In Problem 9 one studies
$$\mathrm{Spec}(A[x]) \to \mathrm{Spec}(A).$$
The fibers are spectra of rings $k(\mathfrak{p})[x]$, which have dimension 1. This fiberwise description is the key reason behind the inequality
$$1 + \dim A \leq \dim A[x] \leq 1 + 2\dim A.$$

19. Dictionary

Ring map $A \to B$	Geometry on spectra
$A \to A/I$	closed immersion $\mathrm{Spec}(A/I) \hookrightarrow \mathrm{Spec}(A)$
$A \to A_f$	open immersion $D(f) \hookrightarrow \mathrm{Spec}(A)$
$A \to A_{\mathfrak{p}}$	local neighborhood of $\mathfrak{p}$
$A \to A[x]$	affine-line type projection
$A \to B$ integral	lying over / going up / surjective on spectra
$A \subseteq B$ domains	dominant map

Problem 1 — Hilbert polynomials of hypersurfaces and complete intersections

Exact statement. Let

$$S = k[x_0, \dots, x_n]$$

be the polynomial ring viewed in the standard way as a graded ring.

(a) Let $f \in S_d$ be a homogeneous polynomial of degree d. Use the exact sequence

$$0 \longrightarrow S(-d) \xrightarrow{\cdot f} S \longrightarrow S/(f) \longrightarrow 0$$

to compute the Hilbert polynomial of the graded S-module $S/(f)$.

(b) Let $f \in S_d$, $g \in S_e$ be two homogeneous polynomials which are coprime. Using a similar idea as in (a), calculate the Hilbert polynomial of $S/(f, g)$.

Solution. Recall that for

$$S = k[x_0, \dots, x_n],$$

the Hilbert function is

$$H_S(m) = \dim_k S_m = \binom{m+n}{n},$$

so the Hilbert polynomial is

$$P_S(t) = \binom{t+n}{n}.$$

Also, for a graded shift $S(-d)$,

$$(S(-d))_m = S_{m-d},$$

hence

$$P_{S(-d)}(t) = \binom{t-d+n}{n}.$$

(a) Hilbert polynomial of $S/(f)$.
From the exact sequence

$$0 \longrightarrow S(-d) \xrightarrow{\cdot f} S \longrightarrow S/(f) \longrightarrow 0,$$

taking graded pieces yields

$$0 \longrightarrow S_{m-d} \xrightarrow{\cdot f} S_m \longrightarrow (S/(f))_m \longrightarrow 0.$$

Therefore
$$H_{S/(f)}(m) = \binom{m+n}{n} - \binom{m-d+n}{n}$$
for all sufficiently large m. Hence the Hilbert polynomial is
$$P_{S/(f)}(t) = \binom{t+n}{n} - \binom{t-d+n}{n}.$$

Its leading term is
$$\frac{d}{(n-1)!} t^{n-1},$$
so it has degree $n-1$.

If $n = 2$, then
$$P_{S/(f)}(t) = \binom{t+2}{2} - \binom{t-d+2}{2} = dt + 1 - \frac{(d-1)(d-2)}{2}.$$
For a plane curve, the constant term is $1 - p_a$, where p_a is the arithmetic genus.

(b) Hilbert polynomial of $S/(f,g)$.

Since f and g are coprime, multiplication by g on $S/(f)$ is injective: if $gh \in (f)$, then $f \mid gh$, and since $\gcd(f,g) = 1$ in the UFD S, it follows that $f \mid h$, so $h \in (f)$.

Thus we have an exact sequence
$$0 \longrightarrow S/(f)(-e) \xrightarrow{\cdot g} S/(f) \longrightarrow S/(f,g) \longrightarrow 0.$$
Hence
$$P_{S/(f,g)}(t) = P_{S/(f)}(t) - P_{S/(f)}(t-e).$$
Using part (a),
$$P_{S/(f,g)}(t) = \binom{t+n}{n} - \binom{t-d+n}{n} - \binom{t-e+n}{n} + \binom{t-d-e+n}{n}.$$

Its leading term is
$$\frac{de}{(n-2)!} t^{n-2},$$
so it has degree $n-2$.

Problem 2 — Degree from the Hilbert polynomial and Bezout-type multiplication

Exact statement. Continue with the notation as in Question 1. Let $I \subseteq S$ be a homogeneous ideal, and let r be the degree of the Hilbert polynomial $f_{S/I}(x)$ of the graded S-module S/I. We define the degree of I to be the coefficient of x^r in $f_{S/I}(x)$ times $r!$.

(a) For $I = (f)$ or $I = (f,g)$ as in Question 1, calculate the degree of I.

(b) Let I be of degree d, and let $f \in S$ be a homogeneous polynomial of degree e such that f is not a zero-divisor in S/I. Show that the degree of $I + (f)$ is e times the degree of I.

Solution.
(a) From Problem 1(a),

$$P_{S/(f)}(t) = \binom{t+n}{n} - \binom{t-d+n}{n},$$

whose leading term is

$$\frac{d}{(n-1)!}t^{n-1}.$$

Therefore

$$\deg(f) = d.$$

From Problem 1(b),

$$P_{S/(f,g)}(t) = \binom{t+n}{n} - \binom{t-d+n}{n} - \binom{t-e+n}{n} + \binom{t-d-e+n}{n},$$

whose leading term is

$$\frac{de}{(n-2)!}t^{n-2}.$$

Therefore

$$\deg(f,g) = de.$$

(b) Let

$$P_{S/I}(t) = a_r t^r + \text{lower degree terms},$$

so by definition

$$\deg(I) = r!\, a_r = d.$$

Since f has degree e and is not a zero-divisor in S/I, multiplication by f gives an injective graded map

$$S/I(-e) \xrightarrow{\cdot f} S/I,$$

and hence an exact sequence

$$0 \longrightarrow S/I(-e) \xrightarrow{\cdot f} S/I \longrightarrow S/(I+(f)) \longrightarrow 0.$$

Therefore

$$P_{S/(I+(f))}(t) = P_{S/I}(t) - P_{S/I}(t-e).$$

Now

$$P_{S/I}(t-e) = a_r(t-e)^r + \cdots = a_r\big(t^r - re\,t^{r-1} + \cdots\big) + \cdots,$$

so

$$P_{S/I}(t) - P_{S/I}(t-e) = a_r re\, t^{r-1} + \text{lower degree terms}.$$

Thus the leading coefficient of $P_{S/(I+(f))}$ is

$$a_r re = \frac{d}{r!}\, re = \frac{ed}{(r-1)!}.$$

Hence

$$\deg(I+(f)) = ed = e\,\deg(I).$$

Problem 3 — Residue fields and localization of graded pieces

Exact statement. Let A be a ring, $p \subseteq A$ a prime ideal. Show that

$$A_p/pA_p$$

is the field of fractions of A/p. Thus in particular, if $m \subseteq A$ is maximal, then

$$A/m \cong A_m/mA_m.$$

Now show that

$$m^k/m^{k+1} \cong (mA_m)^k/(mA_m)^{k+1}.$$

Solution. Let

$$S = A \setminus p.$$

Then

$$A_p = S^{-1}A.$$

Since localization commutes with quotients,

$$A_p/pA_p \cong S^{-1}(A/p).$$

Because p is prime, A/p is a domain. The image of S in A/p is exactly the set of all nonzero elements, so

$$S^{-1}(A/p)$$

is the localization of the domain A/p at all nonzero elements, i.e. its field of fractions. Hence

$$A_p/pA_p \cong \mathrm{Frac}(A/p).$$

If m is maximal, then A/m is already a field, so

$$A_m/mA_m \cong A/m.$$

Now start from the exact sequence

$$0 \longrightarrow m^{k+1} \longrightarrow m^k \longrightarrow m^k/m^{k+1} \longrightarrow 0.$$

Localizing at $A \setminus m$ gives

$$0 \longrightarrow (m^{k+1})_m \longrightarrow (m^k)_m \longrightarrow (m^k/m^{k+1})_m \longrightarrow 0.$$

But

$$(m^k)_m = (mA_m)^k, \qquad (m^{k+1})_m = (mA_m)^{k+1},$$

so

$$(m^k/m^{k+1})_m \cong (mA_m)^k/(mA_m)^{k+1}.$$

It remains to show that localization does not change m^k/m^{k+1}. Since

$$m \cdot m^k \subseteq m^{k+1},$$

the module m^k/m^{k+1} is annihilated by m, hence is naturally an A/m-vector space. Every $s \in A \setminus m$ maps to a nonzero element of the field A/m, hence acts invertibly on m^k/m^{k+1}. Therefore

$$(m^k/m^{k+1})_m \cong m^k/m^{k+1}.$$

Combining the isomorphisms yields

$$m^k/m^{k+1} \cong (mA_m)^k/(mA_m)^{k+1}.$$

Problem 4 — The associated graded ring of an ideal

Exact statement. Let A be a Noetherian ring, $I \subseteq A$ an ideal, and

$$\mathrm{gr}_I(A) = \bigoplus_{n=0}^{\infty} I^n/I^{n+1}$$

the graded ring defined by I.

(a) Show that $\mathrm{gr}_I(A)$ is Noetherian.

(b) Suppose further that A is a local ring, $I = m$ the unique maximal ideal. Show that if $\mathrm{gr}_m(A)$ is an integral domain, then so is A.

Solution.
(a) Since A is Noetherian, the ideal I is finitely generated:

$$I = (a_1, \dots, a_r).$$

Let $\overline{a_i}$ denote the image of a_i in

$$I/I^2 \subseteq \mathrm{gr}_I(A).$$

We claim that $\mathrm{gr}_I(A)$ is generated as an A/I-algebra by $\overline{a_1}, \dots, \overline{a_r}$.

Indeed, every element of I^n/I^{n+1} is represented by an element of I^n, and every element of I^n is a sum of terms

$$c\, a_{i_1} \cdots a_{i_n}, \qquad c \in A.$$

Modulo I^{n+1}, only the class of c modulo I matters. Hence every homogeneous element of degree n in $\mathrm{gr}_I(A)$ is a polynomial in the $\overline{a_i}$ with coefficients in A/I.

Therefore there is a surjective graded homomorphism

$$(A/I)[T_1, \dots, T_r] \twoheadrightarrow \mathrm{gr}_I(A), \qquad T_i \mapsto \overline{a_i}.$$

Since A/I is Noetherian, the polynomial ring $(A/I)[T_1, \dots, T_r]$ is Noetherian, and hence so is its quotient $\mathrm{gr}_I(A)$.

(b) Assume now that A is Noetherian local with maximal ideal m, and that $\mathrm{gr}_m(A)$ is an integral domain.

Take nonzero $a, b \in A$. By the Krull intersection theorem,

$$\bigcap_{r \geq 0} m^r = 0.$$

Hence there exist integers $r, s \geq 0$ such that

$$a \in m^r \setminus m^{r+1}, \qquad b \in m^s \setminus m^{s+1}.$$

Let

$$a^* \in m^r/m^{r+1}, \qquad b^* \in m^s/m^{s+1}$$

be their initial forms in $\mathrm{gr}_m(A)$. These are nonzero.

Their product is the class of ab in

$$m^{r+s}/m^{r+s+1}.$$

If $ab = 0$, then this class is zero, so

$$a^*b^* = 0$$

in $\mathrm{gr}_m(A)$. Since $\mathrm{gr}_m(A)$ is a domain and $a^*, b^* \neq 0$, this is impossible.

Therefore $ab \neq 0$ whenever $a, b \neq 0$. Hence A is an integral domain.

Problem 5 — Jacobian criterion for regularity at the origin

Exact statement. Let

$$I = (f_1, \ldots, f_m) \subseteq k[x_1, \ldots, x_n]$$

be an ideal contained in the maximal ideal

$$\mathfrak{m} = (x_1, \ldots, x_n).$$

Let

$$A = \big(k[x_1, \ldots, x_n]/I\big)_{\bar{\mathfrak{m}}}, \qquad \bar{\mathfrak{m}} = \mathfrak{m}/I.$$

Show that A is a regular local ring if and only if the Jacobian matrix

$$\left(\left.\frac{\partial f_i}{\partial x_j}\right|_{x_1=\cdots=x_n=0} \right)_{1\le i\le m,\ 1\le j\le n}$$

has rank $n-d$, where $d = \dim A$.

Solution. Set

$$R = k[x_1, \ldots, x_n]_{\mathfrak{m}}, \qquad \mathfrak{n} = (x_1, \ldots, x_n)R.$$

Then

$$A = R/IR,$$

and the maximal ideal of A is

$$\mathfrak{m}_A = \mathfrak{n}/IR.$$

A Noetherian local ring $(B, \mathfrak{m}_B)$ is regular if and only if

$$\dim_{B/\mathfrak{m}_B}(\mathfrak{m}_B/\mathfrak{m}_B^2) = \dim B.$$

So we must compute the dimension of

$$\mathfrak{m}_A/\mathfrak{m}_A^2.$$

Now

$$\mathfrak{m}_A/\mathfrak{m}_A^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + IR).$$

Since localization at $\mathfrak{m}$ does not change this k-vector-space computation, we may identify this with

$$\mathfrak{m}/(\mathfrak{m}^2 + I).$$

Also,

$$\mathfrak{m}/\mathfrak{m}^2$$

has k-basis given by the classes of $x_1, \ldots, x_n$, so

$$\dim_k(\mathfrak{m}/\mathfrak{m}^2) = n.$$

We now determine the image of I in $\mathfrak{m}/\mathfrak{m}^2$. For any polynomial $f \in I \subseteq \mathfrak{m}$, modulo $\mathfrak{m}^2$ only its linear part survives:

$$f \equiv \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(0)\, x_j \pmod{\mathfrak{m}^2}.$$

Hence for each generator f_i,

$$f_i \bmod \mathfrak{m}^2 = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}\bigg|_0 x_j.$$

Therefore the image of I in $\mathfrak{m}/\mathfrak{m}^2$ is the k-subspace spanned by the linear forms corresponding to the rows of the Jacobian matrix

$$J(0) = \left(\frac{\partial f_i}{\partial x_j}\bigg|_0 \right).$$

So

$$\dim_k \frac{\mathfrak{m}_A}{\mathfrak{m}_A^2} = \dim_k \frac{\mathfrak{m}}{\mathfrak{m}^2 + I} = n - \operatorname{rank} J(0).$$

Let

$$d = \dim A.$$

Then A is regular if and only if

$$\dim_k(\mathfrak{m}_A/\mathfrak{m}_A^2) = d.$$

Using the formula above, this is equivalent to

$$n - \operatorname{rank} J(0) = d,$$

that is,

$$\operatorname{rank} J(0) = n - d.$$

Thus A is a regular local ring if and only if the Jacobian matrix at the origin has rank $n - d$.

Geometric interpretation. The space

$$\mathfrak{m}_A/\mathfrak{m}_A^2$$

is the cotangent space at the point $\bar{\mathfrak{m}}$, and its dual is the Zariski tangent space. The Jacobian matrix gives the linear equations defining the tangent space, so

$$\dim(\text{tangent space}) = n - \operatorname{rank} J(0).$$

Regularity means that this tangent-space dimension equals the local dimension d.

Theory for Problem 5: local rings, regular local rings, and the Jacobian criterion

1. Why the problem is local

We consider

$$A = \big(k[x_1, \dots, x_n]/I\big)_{\bar{\mathfrak{m}}}, \qquad \bar{\mathfrak{m}} = (x_1, \dots, x_n)/I.$$

This means that we first form the quotient ring $k[x_1, \dots, x_n]/I$, and then localize at the point corresponding to the origin. Thus the problem is not global: it studies the geometry only near one chosen point.

2. Local rings

A ring A is called *local* if it has exactly one maximal ideal. One writes a local ring as

$$(A, \mathfrak{m}),$$

where $\mathfrak{m}$ is the unique maximal ideal.

Examples:

- a field k, whose only maximal ideal is (0);
- a localization $R_{\mathfrak{p}}$, whose unique maximal ideal is $\mathfrak{p}R_{\mathfrak{p}}$;
- the ring
$$\big(k[x_1, \dots, x_n]/I\big)_{\bar{\mathfrak{m}}}$$
appearing in the problem.

Geometrically, a local ring describes the behavior of a space near one point.

3. Residue field

If $(A, \mathfrak{m})$ is local, then

$$A/\mathfrak{m}$$

is a field, called the *residue field*. In the present setting, at the origin this is just k, so tangent-space computations become linear algebra over k.

4. The maximal ideal and first-order information

In a local ring $(A, \mathfrak{m})$,

- $\mathfrak{m}$ consists of functions vanishing at the point;
- $\mathfrak{m}^2$ consists of functions vanishing to second order or higher.

Therefore the quotient

$$\mathfrak{m}/\mathfrak{m}^2$$

records only first-order information. It is called the *cotangent space*. Its dual

$$(\mathfrak{m}/\mathfrak{m}^2)^{\vee}$$

is the *Zariski tangent space*.

5. Embedding dimension

The number

$$\operatorname{embdim}(A) := \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$$

is called the *embedding dimension* of the local ring A. It is the dimension of the tangent space.

For example, in

$$k[x_1, \dots, x_n]_{(x_1, \dots, x_n)},$$

the maximal ideal is generated by $x_1, \dots, x_n$, and their classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$, so the embedding dimension is n.

6. Krull dimension versus embedding dimension

For a Noetherian local ring $(A, \mathfrak{m})$, one always has

$$\dim A \leq \operatorname{embdim}(A).$$

The left-hand side is the intrinsic geometric dimension, while the right-hand side is the tangent-space dimension. At a singular point, the tangent space is too large.

7. Regular local rings

A Noetherian local ring $(A, \mathfrak{m})$ is called *regular* if

$$\dim A = \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2).$$

Equivalently,

$$\dim A = \operatorname{embdim}(A).$$

So a regular local ring is one for which the tangent-space dimension is exactly the true local dimension. This is the algebraic formulation of smoothness or nonsingularity.

8. Regular system of parameters

If $(A, \mathfrak{m})$ is regular of dimension d, then one can choose elements

$$t_1, \dots, t_d \in \mathfrak{m}$$

whose images form a basis of $\mathfrak{m}/\mathfrak{m}^2$. These are called a *regular system of parameters.* They play the role of local coordinates.

9. The basic model

The standard example of a regular local ring is

$$k[x_1, \dots, x_d]_{(x_1,\dots,x_d)}.$$

Its maximal ideal is $(x_1, \dots, x_d)$, the quotient $\mathfrak{m}/\mathfrak{m}^2$ has basis $x_1, \dots, x_d$, and the Krull dimension is d. Hence it is regular.

10. Linear parts and the Jacobian matrix

Suppose

$$I = (f_1, \dots, f_m) \subseteq k[x_1, \dots, x_n].$$

Modulo

$$(x_1, \dots, x_n)^2,$$

only the linear part of each f_i survives:

$$f_i \equiv \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(0)\, x_j \pmod{(x_1, \dots, x_n)^2}.$$

Thus the Jacobian matrix at the origin,

$$J(0) = \left(\frac{\partial f_i}{\partial x_j}(0) \right),$$

records the linear parts of the defining equations.

11. Tangent space from the Jacobian

The ambient affine space has tangent dimension n. Each independent linear equation cuts this dimension down by 1. Therefore

$$\dim T = n - \operatorname{rank} J(0).$$

This equals

$$\dim_k(\mathfrak{m}_A/\mathfrak{m}_A^2),$$

the embedding dimension of the local ring A.

Hence, if $d = \dim A$, then

$$A \text{ is regular} \iff \dim_k(\mathfrak{m}_A/\mathfrak{m}_A^2) = d \iff n - \operatorname{rank} J(0) = d \iff \operatorname{rank} J(0) = n - d.$$

12. Example of a regular point

Let

$$A = \big(k[x,y]/(y-x^2)\big)_{(x,y)}.$$

Then

$$f = y - x^2, \qquad \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (-2x, 1).$$

At the origin this becomes

$$(0,1),$$

so the Jacobian rank is 1. Since $n = 2$ and $\dim A = 1$, one has

$$1 = 2 - 1,$$

so the local ring is regular.

13. Example of a singular point

Let

$$A = \big(k[x,y]/(y^2-x^3)\big)_{(x,y)}.$$

Then

$$f = y^2 - x^3, \qquad \frac{\partial f}{\partial x} = -3x^2, \qquad \frac{\partial f}{\partial y} = 2y.$$

At the origin both derivatives vanish, so the Jacobian rank is 0. Therefore

$$\dim_k(\mathfrak{m}_A/\mathfrak{m}_A^2) = 2.$$

But the local dimension is 1, so the ring is not regular.

14. Simple rings

A ring A is called *simple* if its only ideals are

$$(0) \quad \text{and} \quad A.$$

In the commutative case, a simple ring is exactly a field. Thus in commutative algebra:

- simple ring = field;
- a field is local, with maximal ideal (0);
- a field is regular of dimension 0.

15. Summary

The local ring

$$A = \big(k[x_1, \dots, x_n]/I\big)_{\bar{\mathfrak{m}}}$$

describes the variety near the origin. Its maximal ideal records functions vanishing at the point, and the quotient

$$\mathfrak{m}_A/\mathfrak{m}_A^2$$

records first-order behavior. The Jacobian matrix gives the linear parts of the defining equations, so it computes the tangent-space dimension:

$$\dim_k(\mathfrak{m}_A/\mathfrak{m}_A^2) = n - \operatorname{rank} J(0).$$

The ring is regular exactly when this equals the local dimension d, that is,

$$\operatorname{rank} J(0) = n - d.$$

Deep theory: tangent space, cotangent space, smoothness, singular points, and the Jacobian criterion

1. Why tangent space appears in algebra

In geometry, one studies a space near a point by replacing it with its first-order linear approximation: tangent line, tangent plane, or tangent affine subspace. In commutative algebra, this first-order information is encoded by

$$\mathfrak{m}/\mathfrak{m}^2,$$

where $(A, \mathfrak{m})$ is the local ring at the point.

Thus the Jacobian criterion expresses the principle that a point is smooth exactly when its first-order approximation has the correct dimension.

2. Vanishing at a point

Let $(A, \mathfrak{m})$ be a local ring corresponding to a point x. Then:

- elements of A are local functions near x;
- elements of $\mathfrak{m}$ are functions vanishing at x;
- elements of $\mathfrak{m}^2$ vanish to second order or higher.

Hence the quotient

$$\mathfrak{m}/\mathfrak{m}^2$$

retains only the first-order part of vanishing functions.

3. Cotangent space

If $k(x) = A/\mathfrak{m}$ is the residue field, then

$$\mathfrak{m}/\mathfrak{m}^2$$

is a finite-dimensional $k(x)$-vector space called the *cotangent space* at x.

4. Tangent space

The *Zariski tangent space* at x is defined as

$$T_xX := \mathrm{Hom}_{k(x)}(\mathfrak{m}/\mathfrak{m}^2,\ k(x)).$$

Therefore

$$\dim T_xX = \dim_{k(x)}(\mathfrak{m}/\mathfrak{m}^2).$$

So the dimensions of tangent and cotangent space are the same.

5. Tangent space in affine space

For

$$A = k[x_1, \dots, x_n]_{(x_1,\dots,x_n)},$$

the maximal ideal is

$$\mathfrak{m} = (x_1, \dots, x_n),$$

and the classes of $x_1, \dots, x_n$ form a basis of

$$\mathfrak{m}/\mathfrak{m}^2.$$

Hence

$$\dim_k(\mathfrak{m}/\mathfrak{m}^2) = n,$$

so the tangent space of $\mathbb{A}^n$ at the origin has dimension n.

6. Tangent space of a quotient

Let

$$A = \big(k[x_1, \dots, x_n]/I\big)_{\bar{\mathfrak{m}}}, \qquad \bar{\mathfrak{m}} = (x_1, \dots, x_n)/I.$$

Then

$$\mathfrak{m}_A/\mathfrak{m}_A^2 \cong (x_1, \dots, x_n)/(I + (x_1, \dots, x_n)^2).$$

Thus the cotangent space is obtained from the ambient first-order directions by quotienting out the linear parts of the equations in I.

7. Linearization and the Jacobian matrix

If

$$f \in (x_1, \dots, x_n),$$

then modulo

$$(x_1, \dots, x_n)^2,$$

only the linear part survives:

$$f \equiv \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(0)\, x_j \pmod{(x_1, \dots, x_n)^2}.$$

Therefore the Jacobian matrix

$$J(0) = \left(\frac{\partial f_i}{\partial x_j}(0)\right)$$

records the linear equations defining the tangent space.

8. Smooth and singular points

Let $(A, \mathfrak{m})$ be a Noetherian local ring. The point is called *regular* or *nonsingular* if

$$\dim A = \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2).$$

Equivalently, the tangent-space dimension equals the intrinsic local dimension.

If instead

$$\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) > \dim A,$$

then the tangent space is too large, and the point is singular.

9. Tangent-space dimension from the Jacobian

Suppose

$$X = V(f_1, \ldots, f_m) \subseteq \mathbb{A}^n_k$$

and x is a k-rational point. Then

$$T_xX = \ker\big(J(x) : k^n \to k^m\big),$$

so

$$\dim T_xX = n - \operatorname{rank} J(x).$$

If the local dimension at x is d, then

$$x \text{ is smooth} \iff \dim T_xX = d \iff \operatorname{rank} J(x) = n - d.$$

This is the geometric Jacobian criterion.

10. Example: smooth parabola

Let

$$X = V(y - x^2) \subseteq \mathbb{A}^2.$$

Then

$$f = y - x^2, \qquad \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (-2x, 1).$$

At the origin this becomes

$$(0, 1),$$

so the Jacobian rank is 1. Hence

$$\dim T_{(0,0)}X = 2 - 1 = 1.$$

Since the local dimension is also 1, the origin is smooth.

11. Example: cusp

Let

$$X = V(y^2 - x^3) \subseteq \mathbb{A}^2.$$

Then

$$\frac{\partial f}{\partial x} = -3x^2, \qquad \frac{\partial f}{\partial y} = 2y.$$

At the origin both vanish, so the Jacobian rank is 0. Hence

$$\dim T_{(0,0)}X = 2.$$

But the curve has local dimension 1, so the origin is singular.

12. Example: node

Let

$$X = V(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2.$$

Again, all partial derivatives vanish at the origin, so the Jacobian rank is 0, the tangent space has dimension 2, and the point is singular.

13. Complete intersections and expected codimension

If near a point a variety inside $\mathbb{A}^n$ is cut out by r independent equations, then one expects local dimension

$$n - r.$$

The Jacobian rank counts how many independent linear equations survive to first order. Thus the tangent space has dimension

$$n - r$$

when the point is smooth.

14. Relation with Kähler differentials

A deeper viewpoint uses the module of differentials $\Omega_{A/k}$. At a point x, the cotangent space is closely related to

$$\Omega_{A/k} \otimes_A k(x).$$

Thus the Jacobian criterion can also be understood as saying that the differentials have the correct rank.

15. Regular local rings as local models of smoothness

A regular local ring behaves, to first order, like a polynomial ring in exactly $\dim A$ variables. This is why regular local rings are the algebraic local models of smooth points.

16. Regular versus smooth

Regularity is an intrinsic property of a local ring. Smoothness is a property relative to a base field or base ring. Over a perfect field and for finitely generated algebras, regular points and smooth points coincide. This is the standard setting of the Jacobian criterion.

17. Final picture

For

$$A = \big(k[x_1, \ldots, x_n]/I\big)_{\bar{\mathfrak{m}}},$$

the maximal ideal records functions vanishing at the point, and

$$\mathfrak{m}_A/\mathfrak{m}_A^2$$

records first-order behavior. The Jacobian matrix gives the linear parts of the defining equations, so

$$\dim_k(\mathfrak{m}_A/\mathfrak{m}_A^2) = n - \operatorname{rank} J(x).$$

The local ring is regular exactly when this equals the local dimension d, that is,

$$\operatorname{rank} J(x) = n - d.$$

Example sheet: Jacobian criterion in practice

We use the rule: if

$$X = V(f_1, \dots, f_m) \subseteq \mathbb{A}^n_k$$

and $x \in X$, then

$$\dim T_x X = n - \operatorname{rank} J(x),$$

where

$$J(x) = \left(\frac{\partial f_i}{\partial x_j}(x) \right).$$

If the local dimension at x is d, then x is regular if and only if

$$\operatorname{rank} J(x) = n - d.$$

Example 1 — Smooth parabola at the origin

Let

$$X = V(y - x^2) \subseteq \mathbb{A}^2.$$

Then

$$f = y - x^2, \qquad \frac{\partial f}{\partial x} = -2x, \qquad \frac{\partial f}{\partial y} = 1.$$

At $(0,0)$,

$$J(0,0) = (0 \ \ 1),$$

so

$$\operatorname{rank} J(0,0) = 1.$$

Hence

$$\dim T_{(0,0)} X = 2 - 1 = 1.$$

Since X is a curve, its local dimension is 1. Therefore the origin is smooth.

Example 2 — Cusp

Let

$$X = V(y^2 - x^3) \subseteq \mathbb{A}^2.$$

Then

$$f = y^2 - x^3, \qquad \frac{\partial f}{\partial x} = -3x^2, \qquad \frac{\partial f}{\partial y} = 2y.$$

At $(0,0)$,

$$J(0,0) = (0 \ \ 0),$$

so

$$\operatorname{rank} J(0,0) = 0, \qquad \dim T_{(0,0)} X = 2.$$

But the local dimension is 1. Therefore the origin is singular.

Example 3 — Node

Let

$$X = V(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2.$$

Then

$$f = y^2 - x^2 - x^3, \qquad \frac{\partial f}{\partial x} = -2x - 3x^2, \qquad \frac{\partial f}{\partial y} = 2y.$$

At $(0,0)$,

$$J(0,0) = (0 \;\; 0),$$

so

$$\operatorname{rank} J(0,0) = 0, \qquad \dim T_{(0,0)}X = 2.$$

Since the local dimension is 1, the origin is singular.

Example 4 — Double line

Let

$$X = V(y^2) \subseteq \mathbb{A}^2.$$

Then

$$f = y^2, \qquad \frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 2y.$$

At $(0,0)$,

$$J(0,0) = (0 \;\; 0),$$

so

$$\operatorname{rank} J(0,0) = 0, \qquad \dim T_{(0,0)}X = 2.$$

But the support has local dimension 1, so the origin is singular.

Example 5 — Quadric cone

Let

$$X = V(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3.$$

Then

$$f = x^2 + y^2 - z^2, \qquad \frac{\partial f}{\partial x} = 2x, \qquad \frac{\partial f}{\partial y} = 2y, \qquad \frac{\partial f}{\partial z} = -2z.$$

At the origin,

$$J(0,0,0) = (0 \;\; 0 \;\; 0),$$

so

$$\operatorname{rank} J(0) = 0, \qquad \dim T_0 X = 3.$$

Since X is a hypersurface in $\mathbb{A}^3$, its local dimension is 2. Hence the origin is singular.

Example 6 — Smooth complete intersection

Let
$$X = V(x,\ y) \subseteq \mathbb{A}^3.$$
Then
$$f_1 = x, \qquad f_2 = y.$$
So
$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$
At every point of X, the rank is 2. Hence
$$\dim T_x X = 3 - 2 = 1.$$
Since X has dimension 1, every point is smooth.

Example 7 — Singular complete intersection

Let
$$X = V(x^2,\ y) \subseteq \mathbb{A}^3.$$
Then
$$f_1 = x^2, \qquad f_2 = y.$$
So
$$J = \begin{pmatrix} 2x & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$
At the origin,
$$J(0,0,0) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$
so
$$\operatorname{rank} J(0) = 1, \qquad \dim T_0 X = 3 - 1 = 2.$$
But the local dimension is 1, so the origin is singular.

Example 8 — Smooth point away from the origin

Let
$$X = V(y - x^2) \subseteq \mathbb{A}^2, \qquad p = (1,1).$$
Then
$$\frac{\partial f}{\partial x} = -2x, \qquad \frac{\partial f}{\partial y} = 1.$$
At p,
$$J(1,1) = (-2 \ \ 1),$$
so
$$\operatorname{rank} J(1,1) = 1, \qquad \dim T_p X = 2 - 1 = 1.$$
Since the local dimension is 1, the point is smooth. Its tangent line is
$$-2(x-1) + (y-1) = 0,$$
equivalently
$$y = 2x - 1.$$

Quick comparison table

Example	n	d	rank J	regular?
$V(y-x^2)\subseteq\mathbb{A}^2$ at $(0,0)$	2	1	1	yes
$V(y^2-x^3)\subseteq\mathbb{A}^2$ at $(0,0)$	2	1	0	no
$V(y^2-x^2-x^3)\subseteq\mathbb{A}^2$ at $(0,0)$	2	1	0	no
$V(y^2)\subseteq\mathbb{A}^2$ at $(0,0)$	2	1	0	no
$V(x^2+y^2-z^2)\subseteq\mathbb{A}^3$ at 0	3	2	0	no
$V(x,y)\subseteq\mathbb{A}^3$	3	1	2	yes
$V(x^2,y)\subseteq\mathbb{A}^3$ at 0	3	1	1	no
$V(y-x^2)\subseteq\mathbb{A}^2$ at $(1,1)$	2	1	1	yes

Zariski tangent space on concrete examples

Recall that if X is a scheme, $x\in X$ is a point, and $\mathfrak{m}_x\subseteq\mathcal{O}_{X,x}$ is the maximal ideal of the local ring at x, then the *Zariski tangent space* at x is defined by

$$T_xX := \mathrm{Hom}_{k(x)}\big(\mathfrak{m}_x/\mathfrak{m}_x^2,\ k(x)\big),$$

where

$$k(x)=\mathcal{O}_{X,x}/\mathfrak{m}_x$$

is the residue field of the point x.

Therefore

$$\dim T_xX=\dim_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2).$$

So the tangent space is dual to the cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$, and tangent and cotangent space have the same dimension.

Interpretation. The ideal $\mathfrak{m}_x$ consists of germs of functions vanishing at the point x. The square $\mathfrak{m}_x^2$ consists of functions vanishing to second order or higher. Hence

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

captures only the *first-order* vanishing behavior near the point. Its dual therefore measures first-order directions through the point, i.e. tangent directions.

In concrete affine examples, there are two main ways to compute T_xX:

1. by computing $\mathfrak{m}_x/\mathfrak{m}_x^2$ directly;
2. by linearizing the defining equations using partial derivatives.

We now do several concrete examples carefully.

1. General affine recipe

Let

$$X=V(f_1,\dots,f_r)\subseteq\mathbb{A}^n_k$$

and let

$$x=(a_1,\dots,a_n)\in X(k)$$

be a k-rational point. Then the Zariski tangent space is

$$T_xX = \left\{ (v_1, \dots, v_n) \in k^n : \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(x)\, v_j = 0 \text{ for all } i = 1, \dots, r \right\}.$$

So one evaluates the Jacobian matrix at the point x, and the tangent space is the kernel of that linear map.

Equivalently, if the Jacobian matrix at x is

$$J_x = \left(\frac{\partial f_i}{\partial x_j}(x) \right),$$

then

$$T_xX = \ker(J_x : k^n \to k^r).$$

This is the linear-algebra form of the definition.

2. Example: the affine line $\mathbb{A}^1_k$

Let

$$X = \mathbb{A}^1_k = \operatorname{Spec}(k[t]).$$

Take a k-rational point

$$x = (t - a), \qquad a \in k.$$

Then

$$\mathcal{O}_{X,x} = k[t]_{(t-a)},$$

and its maximal ideal is

$$\mathfrak{m}_x = (t - a).$$

Therefore

$$\mathfrak{m}_x^2 = (t - a)^2,$$

so

$$\mathfrak{m}_x/\mathfrak{m}_x^2 = (t - a)/(t - a)^2.$$

This is a 1-dimensional vector space over the residue field

$$k(x) = k[t]/(t - a) \cong k.$$

Indeed, it is generated by the class of $t - a$.

Hence

$$\dim_k(\mathfrak{m}_x/\mathfrak{m}_x^2) = 1,$$

and thus

$$\dim_k T_xX = 1.$$

So every point of the affine line has 1-dimensional tangent space.

Interpretation. This matches the geometric intuition: a smooth curve has a 1-dimensional tangent space at each smooth point.

3. Example: the affine plane $\mathbb{A}^2_k$

Now let
$$X = \mathbb{A}^2_k = \operatorname{Spec}(k[x, y]).$$
Take a point
$$p = (a, b) \in \mathbb{A}^2_k(k).$$

Then the local ring is
$$\mathcal{O}_{X,p} = k[x, y]_{(x-a,y-b)},$$
with maximal ideal
$$\mathfrak{m}_p = (x - a,\ y - b).$$

Then
$$\mathfrak{m}_p^2 = \big((x - a)^2,\ (x - a)(y - b),\ (y - b)^2\big).$$

So modulo $\mathfrak{m}_p^2$, all quadratic and higher terms vanish, and only the linear terms $x - a$ and $y - b$ remain. Therefore the classes
$$\overline{x - a}, \qquad \overline{y - b}$$
form a basis of $\mathfrak{m}_p/\mathfrak{m}_p^2$. Hence
$$\dim_k(\mathfrak{m}_p/\mathfrak{m}_p^2) = 2.$$

Thus
$$\dim_k T_p\mathbb{A}^2_k = 2.$$

Again this agrees with geometry: the tangent space to the plane at any point is a 2-dimensional plane.

4. Example: a smooth hypersurface

Let
$$X = V(f) \subseteq \mathbb{A}^n_k$$
be a hypersurface. At a point $p \in X(k)$, the tangent space is given by the single linear equation
$$\sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(p)\, v_j = 0.$$

So:

- if the gradient $\nabla f(p) \neq 0$, then this is one nontrivial linear equation, so the tangent space has codimension 1 in k^n, hence dimension $n - 1$;
- if all partial derivatives vanish at p, then the equation becomes $0 = 0$, so the tangent space is all of k^n, which is too large, and this signals a singularity.

We now see this in specific examples.

5. Example: the parabola $y - x^2 = 0$

Let

$$X = V(y - x^2) \subseteq \mathbb{A}^2_k.$$

This is the affine parabola.

Take the point

$$p = (0, 0).$$

5.1. Jacobian computation

Set

$$f(x, y) = y - x^2.$$

Then

$$\frac{\partial f}{\partial x} = -2x, \qquad \frac{\partial f}{\partial y} = 1.$$

At the point $(0, 0)$, these become

$$\frac{\partial f}{\partial x}(0, 0) = 0, \qquad \frac{\partial f}{\partial y}(0, 0) = 1.$$

Hence the tangent vectors $(u, v) \in k^2$ satisfy

$$0 \cdot u + 1 \cdot v = 0,$$

that is,

$$v = 0.$$

So

$$T_{(0,0)}X = \{(u, 0) : u \in k\}.$$

This is the x-axis, which is 1-dimensional.

5.2. Computation via $\mathfrak{m}/\mathfrak{m}^2$

The coordinate ring of X is

$$A = k[x, y]/(y - x^2).$$

At the origin, the maximal ideal is

$$\mathfrak{m} = (x, y) \subseteq A.$$

But in A, we have

$$y = x^2.$$

Since $x^2 \in \mathfrak{m}^2$, it follows that the class of y in $\mathfrak{m}/\mathfrak{m}^2$ is zero. Therefore only the class of x survives, and

$$\mathfrak{m}/\mathfrak{m}^2$$

is 1-dimensional.

Thus

$$\dim T_{(0,0)}X = 1.$$

5.3. General point on the parabola

Now take a general point

$$p = (a, a^2) \in X(k).$$

Then again

$$f(x, y) = y - x^2,$$

so the tangent equation is

$$-2a\,u + v = 0.$$

Hence

$$T_pX = \{(u, v) \in k^2 : v = 2au\}.$$

This is exactly the classical tangent line

$$y - a^2 = 2a(x - a).$$

So at every point, the parabola has 1-dimensional tangent space, as expected for a smooth curve.

6. Example: a line in the plane

Let

$$X = V(y) \subseteq \mathbb{A}^2_k.$$

This is just the x-axis.

Take the origin $p = (0, 0)$. The defining equation is

$$f(x, y) = y.$$

Then

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 1.$$

So the tangent condition is

$$0 \cdot u + 1 \cdot v = 0,$$

that is,

$$v = 0.$$

Hence

$$T_{(0,0)}X = \{(u, 0) : u \in k\}.$$

Thus the tangent space is the line itself, as it should be.

This example shows that for a smooth linear subvariety, the Zariski tangent space agrees exactly with the geometric linear subspace.

7. Example: the union of the coordinate axes $xy = 0$

Now consider

$$X = V(xy) \subseteq \mathbb{A}^2_k.$$

This is the union of the x-axis and the y-axis.

Take the origin

$$p = (0, 0).$$

7.1. Jacobian computation

Set

$$f(x, y) = xy.$$

Then

$$\frac{\partial f}{\partial x} = y, \qquad \frac{\partial f}{\partial y} = x.$$

At the origin,

$$\frac{\partial f}{\partial x}(0,0) = 0, \qquad \frac{\partial f}{\partial y}(0,0) = 0.$$

So the tangent equation becomes

$$0 = 0.$$

Hence every vector $(u, v) \in k^2$ is tangent, so

$$T_{(0,0)}X = k^2.$$

Thus the tangent space has dimension 2.

7.2. Computation via $\mathfrak{m}/\mathfrak{m}^2$

The coordinate ring is

$$A = k[x, y]/(xy).$$

At the origin,

$$\mathfrak{m} = (x, y) \subseteq A.$$

Now the relation $xy = 0$ lies already in $\mathfrak{m}^2$, so it produces no linear relation in

$$\mathfrak{m}/\mathfrak{m}^2.$$

Hence the classes of x and y remain linearly independent modulo $\mathfrak{m}^2$. Therefore

$$\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2.$$

So again

$$\dim_k T_{(0,0)}X = 2.$$

7.3. Interpretation

The variety X is a curve, so one might expect tangent dimension 1 at smooth points. But at the crossing point the tangent space jumps to dimension 2. This indicates that the origin is singular.

Geometrically, two branches cross there, and the Zariski tangent space is the whole ambient plane.

8. Example: the cusp $y^2 - x^3 = 0$

Consider

$$X = V(y^2 - x^3) \subseteq \mathbb{A}^2_k.$$

Take the origin $p = (0, 0)$.

8.1. Jacobian computation

Set
$$f(x,y) = y^2 - x^3.$$
Then
$$\frac{\partial f}{\partial x} = -3x^2, \qquad \frac{\partial f}{\partial y} = 2y.$$

At the origin,
$$\frac{\partial f}{\partial x}(0,0) = 0, \qquad \frac{\partial f}{\partial y}(0,0) = 0.$$

So again the tangent equation is just
$$0 = 0,$$
hence
$$T_{(0,0)}X = k^2.$$

Therefore
$$\dim_k T_{(0,0)}X = 2.$$

8.2. Computation via $\mathfrak{m}/\mathfrak{m}^2$

The coordinate ring is
$$A = k[x,y]/(y^2 - x^3).$$
At the origin,
$$\mathfrak{m} = (x,y) \subseteq A.$$

The relation
$$y^2 = x^3$$
has degree at least 2, so again it lies in $\mathfrak{m}^2$. Therefore it does not impose any linear relation in
$$\mathfrak{m}/\mathfrak{m}^2.$$

So both x and y survive independently, and
$$\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2.$$

Thus
$$\dim_k T_{(0,0)}X = 2.$$

8.3. Interpretation

The cusp is singular. Even though in elementary geometry one often draws a single tangent line at the cusp, the Zariski tangent space is still 2-dimensional. This is because the tangent space is obtained from all first-order relations, and here there is no nontrivial linear relation at the origin.

So the Zariski tangent space detects singularity very effectively.

9. Example: a smooth circle

Let $\mathrm{char}(k) \neq 2$, and consider

$$X = V(x^2 + y^2 - 1) \subseteq \mathbb{A}_k^2.$$

Take the point

$$p = (1, 0) \in X(k).$$

Let

$$f(x, y) = x^2 + y^2 - 1.$$

Then

$$\frac{\partial f}{\partial x} = 2x, \qquad \frac{\partial f}{\partial y} = 2y.$$

At $(1, 0)$,

$$\frac{\partial f}{\partial x}(1, 0) = 2, \qquad \frac{\partial f}{\partial y}(1, 0) = 0.$$

So the tangent equation is

$$2u + 0 \cdot v = 0,$$

hence

$$u = 0.$$

Therefore

$$T_{(1,0)}X = \{(0, v) : v \in k\},$$

which is the vertical line, exactly as in classical geometry.

This is another example of a smooth curve having 1-dimensional tangent space.

10. Example: the plane curve $y^2 = x^2(x + 1)$

Consider

$$X = V(y^2 - x^2(x + 1)) \subseteq \mathbb{A}_k^2.$$

Expanding,

$$f(x, y) = y^2 - x^3 - x^2.$$

At the origin,

$$\frac{\partial f}{\partial x} = -3x^2 - 2x, \qquad \frac{\partial f}{\partial y} = 2y.$$

So at $(0, 0)$,

$$\frac{\partial f}{\partial x}(0, 0) = 0, \qquad \frac{\partial f}{\partial y}(0, 0) = 0.$$

Hence

$$T_{(0,0)}X = k^2.$$

So the origin is singular.

If one looks only at the lowest-degree homogeneous part of the defining equation, one sees

$$y^2 - x^2 = (y - x)(y + x),$$

which suggests two tangent directions. The Zariski tangent space itself is still 2-dimensional, namely the whole ambient plane, but the tangent cone splits into two lines. This illustrates that tangent space and tangent cone are related but not identical notions.

11. Example: the dual numbers

Now take the nonreduced scheme

$$X = \operatorname{Spec}(k[\varepsilon]/(\varepsilon^2)).$$

This scheme has only one point, with local ring

$$A = k[\varepsilon]/(\varepsilon^2),$$

and maximal ideal

$$\mathfrak{m} = (\varepsilon).$$

Since

$$\varepsilon^2 = 0,$$

we have

$$\mathfrak{m}^2 = 0.$$

Therefore

$$\mathfrak{m}/\mathfrak{m}^2 \cong (\varepsilon),$$

which is 1-dimensional over the residue field

$$k(x) = A/\mathfrak{m} \cong k.$$

Hence

$$\dim_k T_x X = 1.$$

Interpretation

This is very important conceptually: topologically X has only one point, but its tangent space is nonzero. So tangent spaces detect infinitesimal thickness and nilpotent structure, not just the set of ordinary points.

This is one of the key reasons the definition via $\mathfrak{m}/\mathfrak{m}^2$ is so powerful in algebraic geometry.

12. Example: a fat point in the plane

Consider

$$X = \operatorname{Spec}(k[x, y]/(x, y)^2).$$

Here

$$(x, y)^2 = (x^2, xy, y^2),$$

so the coordinate ring is

$$A = k[x, y]/(x^2, xy, y^2).$$

This scheme has one point, corresponding to the maximal ideal

$$\mathfrak{m} = (x, y).$$

Since all products of elements of $\mathfrak{m}$ vanish,

$$\mathfrak{m}^2 = 0.$$

Hence

$$\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{m}$$

is generated by the classes of x and y, so it is 2-dimensional over k. Therefore

$$\dim_k T_x X = 2.$$

Again, there is only one underlying point, but the tangent space is 2-dimensional because the point carries nontrivial infinitesimal structure.

13. Example: a non-k-rational point

The use of the residue field $k(x)$ in the definition becomes important when the point is not k-rational.

Let

$$X = \operatorname{Spec}(\mathbb{R}[t]),$$

and take the closed point

$$x = (t^2 + 1).$$

This is a closed point of the affine line over $\mathbb{R}$, but it is not an $\mathbb{R}$-rational point, because

$$k(x) = \mathbb{R}[t]/(t^2 + 1) \cong \mathbb{C}.$$

Now the maximal ideal in the local ring is generated by the class of $t^2 + 1$, and

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

is 1-dimensional over $\mathbb{C}$. Hence

$$\dim_{\mathbb{C}} T_x X = 1.$$

So the tangent space is 1-dimensional over the residue field $\mathbb{C}$, not over the base field $\mathbb{R}$.

Interpretation

This shows why the definition is

$$T_x X = \operatorname{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x))$$

and not always over the base field k: the natural linear structure of the tangent space is over the residue field of the point.

14. Example: affine n-space

More generally, let

$$X = \mathbb{A}_k^n = \operatorname{Spec}(k[x_1, \dots, x_n]),$$

and let

$$p = (a_1, \dots, a_n) \in \mathbb{A}_k^n(k).$$

Then

$$\mathfrak{m}_p = (x_1 - a_1, \dots, x_n - a_n).$$

Modulo $\mathfrak{m}_p^2$, the classes

$$\overline{x_1 - a_1}, \ldots, \overline{x_n - a_n}$$

form a basis of $\mathfrak{m}_p/\mathfrak{m}_p^2$. Therefore

$$\dim_k(\mathfrak{m}_p/\mathfrak{m}_p^2) = n,$$

and hence

$$\dim_k T_p\mathbb{A}_k^n = n.$$

This is exactly what one expects: the tangent space to n-dimensional affine space is n-dimensional.

15. Smoothness criterion via tangent space

These examples illustrate a central general principle.

If X is a variety of dimension d, then at a smooth point x one expects

$$\dim_{k(x)} T_xX = d.$$

If instead

$$\dim_{k(x)} T_xX > d,$$

then x is singular.

This is exactly what we saw:

- on $\mathbb{A}^1$, tangent dimension is 1;
- on $\mathbb{A}^2$, tangent dimension is 2;
- on the parabola, tangent dimension is 1 at every point;
- at the node $xy = 0$, tangent dimension at the origin is 2;
- at the cusp $y^2 - x^3 = 0$, tangent dimension at the origin is 2;
- for nonreduced schemes like $k[\varepsilon]/(\varepsilon^2)$, the tangent space detects infinitesimal thickening.

So the tangent space is one of the most important tools for detecting smooth points, singular points, and infinitesimal structure.

16. Conceptual summary

Let $x \in X$. Then:

- $\mathfrak{m}_x$ consists of functions vanishing at x;
- $\mathfrak{m}_x^2$ consists of products of such functions, hence second-order vanishing;
- $\mathfrak{m}_x/\mathfrak{m}_x^2$ records first-order vanishing data;
- its dual

  $$T_xX = \mathrm{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x))$$

  is the space of first-order directions through the point.

So one may think of

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

as the *cotangent space*, and

$$T_xX$$

as the *tangent space*, i.e. the best linear approximation to the scheme near the point.

17. One-line intuition

A useful slogan is:

$$\boxed{\text{The Zariski tangent space is what remains after keeping only the linear part of the local equations at}}$$

That is exactly why:

- smooth points give the expected tangent dimension;
- singular points give a tangent space that is too large;
- nilpotents create tangent directions even when there is only one underlying point.

What is $\mathbb{A}^1_k$? Is it a polynomial ring or is it $\mathbb{R}$?

The affine line

$$\mathbb{A}^1_k$$

is *not* itself a polynomial and it is also *not* simply the field k or the set $\mathbb{R}$ (unless one is speaking only informally about rational points).

It is the *affine line over the field* k, defined by

$$\mathbb{A}^1_k = \mathrm{Spec}(k[t]).$$

So one must distinguish carefully between the *geometric object* and the *ring of functions on it.*

1. Algebraic point of view

The ring

$$k[t]$$

is the polynomial ring in one variable over k.

The affine line

$$\mathbb{A}^1_k = \mathrm{Spec}(k[t])$$

is the geometric space associated with that ring.

Thus:

$$k[t] = \text{the ring of polynomial functions in one variable},$$

whereas

$$\mathbb{A}^1_k = \text{the geometric object whose coordinate ring is } k[t].$$

So $\mathbb{A}^1_k$ is *not* the polynomial ring itself. Rather, it is the scheme whose coordinate ring is $k[t]$.

This is exactly analogous to the general philosophy of algebraic geometry:

$$\text{geometric space} \longleftrightarrow \text{ring of functions on that space.}$$

In this case,

$$\mathbb{A}^1_k \longleftrightarrow k[t].$$

2. Pointwise point of view

If one asks for the k-rational points of the affine line, then one gets

$$\mathbb{A}^1_k(k) = k.$$

So as far as k-rational points are concerned, the affine line behaves like the set k.

For example:

$$\mathbb{A}^1_{\mathbb{C}}(\mathbb{C}) = \mathbb{C}, \qquad \mathbb{A}^1_{\mathbb{R}}(\mathbb{R}) = \mathbb{R}.$$

This is why people often say informally that "the affine line over k is just k."

But this is only true at the level of k-rational points. As a scheme, $\mathbb{A}^1_k$ contains more structure than just the set k.

3. Why $\mathbb{A}^1_{\mathbb{R}}$ is more than just $\mathbb{R}$

Over $\mathbb{R}$, one has

$$\mathbb{A}^1_{\mathbb{R}} = \mathrm{Spec}(\mathbb{R}[t]).$$

Its $\mathbb{R}$-rational points are indeed the real numbers:

$$\mathbb{A}^1_{\mathbb{R}}(\mathbb{R}) = \mathbb{R}.$$

However, as a scheme, $\mathbb{A}^1_{\mathbb{R}}$ also has closed points that are not real points in the ordinary sense.

For example, the ideal

$$(t^2 + 1)$$

is maximal in $\mathbb{R}[t]$, because $t^2 + 1$ is irreducible over $\mathbb{R}$. So it defines a closed point of $\mathrm{Spec}(\mathbb{R}[t])$.

The residue field of this point is

$$\mathbb{R}[t]/(t^2 + 1) \cong \mathbb{C}.$$

So this point is not an $\mathbb{R}$-rational point; it is a closed point whose field of definition is $\mathbb{C}$.

This shows that

$$\mathbb{A}^1_{\mathbb{R}}$$

as a scheme is richer than the ordinary set $\mathbb{R}$.

4. Closed points of $\mathrm{Spec}(k[t])$

More generally, the points of

$$\mathrm{Spec}(k[t])$$

correspond to prime ideals of $k[t]$.

Since $k[t]$ is a principal ideal domain, its prime ideals are:

- the zero ideal (0), which is the generic point;
- ideals (f), where $f \in k[t]$ is irreducible.

So the closed points of $\mathbb{A}^1_k$ correspond to irreducible polynomials in $k[t]$, not just to elements of k.

If k is algebraically closed, every irreducible polynomial is linear, so every closed point has the form

$$(t-a), \qquad a \in k.$$

In that case the closed points really do correspond exactly to elements of k.

But if k is not algebraically closed, then there are also irreducible polynomials of higher degree, and these give additional closed points.

5. Coordinate ring versus underlying set

It is very important not to confuse:

- the field k,
- the polynomial ring $k[t]$,
- the affine line $\mathbb{A}^1_k = \mathrm{Spec}(k[t])$.

These are three different things.

(a) The field k. This is the base field of scalars.

(b) The ring $k[t]$. This is the ring of polynomial expressions in one variable with coefficients in k.

(c) The scheme $\mathbb{A}^1_k$. This is the geometric object associated to the ring $k[t]$.

So:

$$k \neq k[t] \neq \mathbb{A}^1_k.$$

They are related, but they are not the same object.

6. Analogy

A very useful analogy is:

$$\mathbb{A}^1_k \longleftrightarrow \text{a line}, \qquad k[t] \longleftrightarrow \text{polynomial functions on that line.}$$

So the line itself is the geometric object, and the polynomial ring is the algebra of functions on it.

This is similar to the idea from calculus or differential geometry that one distinguishes between:

- a space,
- and functions defined on that space.

7. Best summary

The cleanest summary is:

$$\boxed{\mathbb{A}^1_k \text{ is the geometric object, and } k[t] \text{ is its polynomial coordinate ring.}}$$

So if one asks:

- "Is $\mathbb{A}^1$ a polynomial ring?" — no. It is the geometric space associated to the polynomial ring $k[t]$.
- "Is $\mathbb{A}^1_{\mathbb{R}}$ equal to $\mathbb{R}$?" — its $\mathbb{R}$-rational points are $\mathbb{R}$, but as a scheme it contains more information than the set $\mathbb{R}$ alone.

8. Final slogan

A useful slogan is:

$$\boxed{\mathbb{A}^1_k = \text{the affine line over } k, \qquad k[t] = \text{the ring of polynomial functions on that line.}}$$

Dual numbers and their geometric meaning

The *dual numbers* over a field k are defined as the quotient ring

$$k[\varepsilon]/(\varepsilon^2).$$

This means that ε is a formal symbol satisfying

$$\varepsilon^2 = 0,$$

but $\varepsilon \neq 0$.

So every element of the dual number ring has the form

$$a + b\varepsilon, \qquad a, b \in k.$$

This looks formally similar to complex numbers $a + bi$, but the rule is different: for complex numbers one has $i^2 = -1$, while for dual numbers one has

$$\varepsilon^2 = 0.$$

Addition and multiplication

If

$$a + b\varepsilon, \qquad c + d\varepsilon \in k[\varepsilon]/(\varepsilon^2),$$

then their sum is

$$(a + b\varepsilon) + (c + d\varepsilon) = (a + c) + (b + d)\varepsilon,$$

and their product is

$$(a+b\varepsilon)(c+d\varepsilon) = ac + (ad+bc)\varepsilon + bd\,\varepsilon^2.$$

Since $\varepsilon^2 = 0$, this simplifies to

$$(a+b\varepsilon)(c+d\varepsilon) = ac + (ad+bc)\varepsilon.$$

So multiplication in the dual numbers keeps only constant and first-order terms.

Concrete example

For example,

$$(2+3\varepsilon)(5+\varepsilon) = 10 + 2\varepsilon + 15\varepsilon + 3\varepsilon^2.$$

Because $\varepsilon^2 = 0$, this becomes

$$(2+3\varepsilon)(5+\varepsilon) = 10 + 17\varepsilon.$$

Units and nilpotents

An element

$$a + b\varepsilon$$

is invertible if and only if $a \neq 0$. Indeed, if $a \neq 0$, then

$$(a+b\varepsilon)^{-1} = a^{-1} - a^{-2}b\,\varepsilon.$$

This is checked directly:

$$(a+b\varepsilon)\big(a^{-1} - a^{-2}b\,\varepsilon\big) = 1 + \big(-a^{-1}b + a^{-1}b\big)\varepsilon = 1.$$

If $a = 0$, then the element is of the form $b\varepsilon$, and

$$(b\varepsilon)^2 = b^2\varepsilon^2 = 0.$$

So such an element is nilpotent.

Thus the dual number ring contains a nonzero nilpotent element, namely ε. This is why it is a standard example of a *nonreduced* ring.

Why dual numbers matter in algebraic geometry

The ring

$$k[\varepsilon]/(\varepsilon^2)$$

is the simplest ring that contains first-order infinitesimal information. The corresponding affine scheme is

$$\operatorname{Spec}\big(k[\varepsilon]/(\varepsilon^2)\big).$$

Topologically, this scheme has only one point, because the only prime ideal is

$$(\varepsilon).$$

Indeed, the quotient by this ideal is

$$k[\varepsilon]/(\varepsilon) \cong k,$$

which is a field, so (ε) is maximal. Since every prime ideal must contain all nilpotent elements, and ε is nilpotent, there are no other prime ideals.

So as a topological space,

$$\operatorname{Spec}\big(k[\varepsilon]/(\varepsilon^2)\big)$$

is just one point.

However, algebraically it is not the same as $\operatorname{Spec}(k)$, because the ring has an extra nilpotent element ε. This means that the point is not an ordinary reduced point, but a *fat point*, or an *infinitesimal thickening* of a point.

Geometric intuition

One may think of an element

$$a + b\varepsilon$$

as consisting of:

- a genuine value a,
- plus a first-order infinitesimal correction $b\varepsilon$.

Because $\varepsilon^2 = 0$, all second-order and higher-order effects disappear. So dual numbers retain exactly first-order information and discard everything quadratic or higher.

This is why they are so naturally connected with tangent vectors and linear approximation.

Dual numbers and tangent vectors

A very important idea in algebraic geometry is that a tangent vector at a point $x \in X$ can be described by a morphism

$$\operatorname{Spec}\big(k[\varepsilon]/(\varepsilon^2)\big) \to X$$

whose closed point maps to x.

Such a morphism represents a first-order infinitesimal motion away from the point x. In other words, it records a direction at x, but only up to first order.

That is why dual numbers are often used as the algebraic object encoding tangent directions.

Relation with the Zariski tangent space

Let

$$X = \operatorname{Spec}\big(k[\varepsilon]/(\varepsilon^2)\big).$$

Since there is only one point, call it x. The local ring at x is

$$\mathcal{O}_{X,x} = k[\varepsilon]/(\varepsilon^2),$$

and its maximal ideal is

$$\mathfrak{m}_x = (\varepsilon).$$

Because

$$\varepsilon^2 = 0,$$

we have

$$\mathfrak{m}_x^2 = 0.$$

Therefore

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong (\varepsilon),$$

which is 1-dimensional over the residue field

$$k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x \cong k.$$

Hence the Zariski tangent space is

$$T_xX = \mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k),$$

so

$$\dim_k T_xX = 1.$$

Thus even though the scheme has only one underlying point, it has a nontrivial tangent direction.

Why this is important

This example shows something fundamental:

> A scheme can have only one point topologically, but still carry nontrivial infinitesimal structure.

That infinitesimal structure is detected by nilpotent elements in the ring, and the dual numbers are the simplest example of this phenomenon.

So dual numbers are not just an algebraic curiosity. They are one of the basic tools for understanding:

- tangent vectors,
- infinitesimal neighborhoods,
- nonreduced schemes,
- first-order deformations.

Summary

The dual numbers over k are

$$k[\varepsilon]/(\varepsilon^2),$$

whose elements are

$$a + b\varepsilon.$$

They satisfy:

- $\varepsilon^2 = 0$,
- $\varepsilon \neq 0$,
- they form a nonreduced ring,

- their spectrum is a single fat point,
- they encode first-order infinitesimal information,
- they are closely tied to tangent spaces and tangent vectors.

A useful slogan is:

$$\boxed{\text{Dual numbers are the algebra of first-order infinitesimals.}}$$